

MATHEMATICS GRADE 10: LINEAR MOTION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.4 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 3.0: Measurements and Geometry
Sub-Strand	Sub-Strand 3.4: Linear Motion
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Distance, displacement, speed, velocity and acceleration in real-life situations – Determining velocity and acceleration in different situations – Drawing displacement-time graphs of different situations – Interpreting displacement-time graphs in different situations – Drawing velocity-time graphs from given tables in different situations – Interpreting velocity-time graphs in different situations – Relative speed of two moving bodies in different situations – Application of linear motion in real-life situations
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - explain the terms distance, displacement, speed, velocity and acceleration in real-life situations - determine velocity and acceleration in different situations - draw displacement-time graphs of different situations - interpret displacement-time graphs in different situations - draw velocity-time graphs from given tables in different situations - interpret velocity-time graphs in different situations - determine the relative speed of two moving bodies in different situations - apply linear motion in real-life situations
Core Competencies	– Critical thinking and problem solving: learners analyse the overtaking video, compare motion quantities and justify conclusions from graphs and calculations. – Communication and collaboration: learners discuss predictions in groups, explain their reasoning using correct terms and present graphs and models to peers. – Digital literacy: learners use video observation and, where available, simulations or graphing tools to collect and represent motion data. – Self-efficacy: learners build confidence by revising their motion model across lessons and solving increasingly challenging problems. – Learning to learn: learners track their questions on the Driving Question Board and reflect on how their understanding has grown.

Core Values	– Responsibility: learners observe road safety rules and take care of shared equipment during measurements. – Integrity: learners record and report measured times and distances honestly, even when results differ from predictions. – Respect: learners listen to and build on peers' ideas during group sensemaking. – Unity: learners work cooperatively in groups to gather data and build shared models.
Science & Engineering Practices	– Asking questions and defining problems: learners generate questions about how one car overtakes another for the Driving Question Board. – Developing and using models: learners build and revise a model of motion using graphs, tables and equations. – Analyzing and interpreting data: learners read displacement-time and velocity-time graphs to determine speed, velocity and acceleration. – Using mathematics and computational thinking: learners calculate velocity, acceleration and relative speed. – Constructing explanations: learners use evidence to explain how and when an overtaking car passes another.
Pertinent & Contemporary Issues (PCIs)	– Road safety education: learners relate speed, overtaking and stopping distances to safe driving on roads such as the Nairobi-Nakuru highway. – Citizenship: learners consider responsible behaviour as road users, including drivers, cyclists and pedestrians. – Safety and security: learners discuss speed limits and the risks of unsafe overtaking.
Career Connections	– Pilot and air traffic controller: use velocity, relative speed and acceleration to manage flight paths. – Automotive engineer and mechanic: design and test vehicles using acceleration and braking data. – Traffic police officer / NTSA officer: apply speed and distance measurements to enforce road safety. – Athletics coach: analyse runners' speed and acceleration, e.g. 800 m races by Kenyan athletes. – Civil engineer: design roads, overtaking lanes and junctions using motion principles.
Focus for Lessons	This unit uses a video of one car overtaking another on a Kenyan road to motivate the study of linear motion. Learners progress from defining distance, displacement, speed, velocity and acceleration to drawing and interpreting displacement-time and velocity-time graphs, then to relative speed and real-life applications. Each lesson follows predict, observe, explain, Driving Question Board and model building phases, so learners keep revising one model of the overtaking event.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we use distance, time, speed, velocity and graphs to describe and predict how one car overtakes another on a Kenyan road? KICD KEY INQUIRY QUESTIONS: 1. How do you determine the velocity of a moving body? 2. How do you determine the acceleration of a moving body?
Anchoring Phenomenon	Learners watch videos of vehicles travelling along a road, such as a matatu, a lorry and private cars on the Nairobi-Nakuru highway near Naivasha, and one clip in which a car overtakes another. Both vehicles are moving in the same direction, yet the overtaking car appears to move quickly past the slower one while the speedometers show speeds that both seem high. Learners notice that the car being overtaken looks as if it is moving slowly from the overtaking driver's point of view, and that the overtaking car must speed up, pass and then return to its lane before an oncoming vehicle arrives. This is puzzling: how can two vehicles that are both moving fast appear to move slowly relative to each other, how long does the overtaking take, how far do the cars travel, and how can we describe and predict the motion using numbers and graphs?
Storyline Thread	Lesson 1: Phenomenon – learners watch videos of vehicles on a road and a car overtaking another, predict what happens, and generate questions for the Driving Question Board.

	Lesson 2: Learners explain distance, displacement, speed, velocity and acceleration using real-life examples such as a boda boda trip and a matatu journey. Lesson 3: Learners determine velocity and acceleration in different situations, including a car speeding up to overtake. Lesson 4: Learners draw displacement-time graphs of different situations, such as a stationary vehicle, a steady car and a returning cyclist. Lesson 5: Learners interpret displacement-time graphs to find velocity, direction and when vehicles meet or are at rest. Lesson 6: Learners draw velocity-time graphs from given tables of data for vehicles speeding up, moving steadily and slowing down. Lesson 7: Learners interpret velocity-time graphs to find acceleration, deceleration and distance travelled from the area under the graph. Lesson 8: Learners determine the relative speed of two bodies moving in the same direction and in opposite directions, explaining why the overtaken car looks slow. Lesson 9: Learners apply linear motion to real-life situations, such as the time and distance needed to overtake safely on a highway. Lesson 10: Learners complete and present their final model of the overtaking event, revisit the Driving Question Board and answer the driving question.
Supporting Phenomena	– A matatu from Kisumu to Nairobi speeds up after leaving a stage, travels steadily on the highway and slows down to stop, so its speedometer reading changes throughout the trip. – A Kenyan 800 m runner such as David Rudisha increases speed at the start, then maintains pace and sprints at the finish. – A boda boda rider travels 3 km away from a shopping centre and then returns, ending close to the starting point although the distance travelled is 6 km. – A Standard Gauge Railway train from Mombasa to Nairobi travels at nearly constant speed on a straight section of track.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (40 minutes (1 lesson)): Phenomenon: Overtaking on the Naivasha Highway

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon by having learners observe a car overtaking another on the Nairobi-Nakuru highway near Naivasha, predict how long and how far the overtake takes, notice what is puzzling, and generate questions for the Driving Question Board.
Knowledge	Learners describe in everyday words what they see in the clip: overtaken car A at 72 km/h, overtaking car B at 72 km/h at the start and then faster, an oncoming lorry L, and car B returning to its lane. They recognise that distance, time, speed, direction and change of speed are the ideas needed to describe the motion. No formulas are taught.
Skills	Asking questions (NGSS SEP 1): learners write testable questions about the overtake. Developing and using models (NGSS SEP 2): learners draw an initial sketch model of the overtake. Analyzing and interpreting data: learners record qualitative observations and make predictions from them. Communicating: learners share ideas in pairs and with the class.
Attitudes	Curiosity about everyday motion on Kenyan roads; respect for others' ideas; willingness to make a prediction and revise it later; a safety-conscious attitude towards road use and overtaking.
Key Inquiry Question	KICD key inquiry questions to be answered over the unit: (1) How do you determine the velocity of a moving body? (2) How do you determine the acceleration of a moving body? In Lesson 1 learners only raise questions that point towards these.
Purpose in Storyline	This lesson launches the storyline. Learners meet the phenomenon and raise questions such as 'how long?', 'how far?', 'how fast?' and 'is it safe?'. These questions open the Driving Question Board and are returned to in Lessons 2 to 10. Lesson 2 builds on the DQB questions about 'how far' and 'how fast'.
Safety Notes	Video is watched indoors or in a shaded classroom; no learner should go onto a real road for this lesson. Stress that the clip is a recorded manoeuvre and that real overtaking is dangerous. Learners must never imitate overtaking on bicycles, boda bodas or on foot near roads. Keep walkways clear when learners move to the DQB.

B. LESSON OVERVIEW

Learners watch short clips of a matatu, a lorry and private cars on the Nairobi-Nakuru highway near Naivasha, filmed by Kiprono from the roadside. They then watch one clip in which Wanjiru in the white Subaru (overtaking car B) passes Mr Kamau's blue Toyota Axio (overtaken car A) while Mr Otieno's lorry (oncoming lorry L) approaches from Nakuru. Before and during the viewing, learners predict how long and how far the overtake will take and sketch what they think happens. Achieng, in the back seat of car B, sees both speedometers reading high, yet car A looks slow from car B. This puzzle is explored through a Notice-Wonder discussion. Learners write questions on sticky notes and build the first Driving Question Board. No formulas, split-time tables or computed values are given in this lesson; the focus is on observing, predicting, puzzling and questioning.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown only a still image of the Nairobi-Nakuru	Display the still image and say: 'Both cars are moving at 72 km/h.'	Predict-Observe-Explain (prediction stage) with Think-Pair-	Look for: (a) predictions that include a time and a distance, not

 VIDEO: Calculating average velocity or speed Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Calculating Acceleration from Velocity and Time (Flexbook) Source: CK-12  Search ARES for similar readings	highway near Naivasha with car A (blue Toyota Axio), car B (white Subaru) just behind it and a lorry far ahead in the opposite lane. They are told that both cars are going about 72 km/h and that Wanjiru in car B will overtake car A. Individually, in their notebooks, they write: (1) a prediction of how long the overtake will take (in seconds: under 3, 3 to 5, 5 to 10, or more than 10); (2) a prediction of how far car B will travel while overtaking; (3) a quick sketch of the positions of A, B and the lorry at the start and the end. They compare with a partner and explain their reasoning using their own experience of travelling in matatus on Kenyan roads.	Wanjiru wants to pass Mr Kamau safely. Before we watch anything, I want your best prediction.' Read the three prompts aloud. Give 3 minutes of silent writing. Say: 'There are no wrong predictions today; we will test them later.' Circulate and note 3 or 4 contrasting predictions. Use WAIT TIME of 10-15 seconds after asking: 'What do you think happens to the speed of car B?' Then cold-call 4 learners to share a prediction and one reason, and record the range of time predictions on the board as tally marks. Do NOT say which prediction is correct and do NOT show numbers beyond 72 km/h.	Share: writing a prediction before seeing evidence makes learners' prior ideas about speed, time and distance visible, so that later evidence can confirm or challenge them. Pairing lets learners rehearse reasoning in low-risk talk before sharing publicly.	just 'fast' or 'slow'; (b) sketches showing car B moving ahead of car A and returning to its lane; (c) sketches that include the oncoming lorry; (d) reasons that mention speeding up. Note on a checklist learners who write only 'it is fast' with no number. Common ideas to record: 'B must go much faster than 72 km/h'; 'the overtake takes about 2 seconds'; 'both cars travel the same distance'.
Observe Phase  VIDEO: Calculating average velocity or speed Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Acceleration and velocity Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners watch two sets of clips. Set 1 (about 3 minutes): vehicles travelling along the Nairobi-Nakuru highway near Naivasha, including a matatu, a lorry and private cars, filmed by Kiprono. Learners complete a Notice column in a two-column table (Notice Wonder) by listing what they see. Set 2 (about 2 minutes, shown twice): the clip of car B overtaking car A while oncoming lorry L approaches. In the clip both cars show about 72 km/h at the start, then car B speeds up, passes car A and returns to its lane before the lorry arrives. Achieng's view from the back seat of car B shows both speedometers reading high, yet car A seems to move slowly relative to car B. Learners record qualitative observations only: who speeds up, who keeps steady speed, direction of each vehicle, the moment B pulls back into lane. They compare what they saw with their predictions.	Introduce: 'Kiprono stood at the roadside near Naivasha and filmed these vehicles. Watch like a scientist and write what you see, not what you think.' Play Set 1 once without commentary. Play Set 2 first with no pause, then say: 'Watch again. This time focus on the speedometers and on how car A looks from car B.' Pause at the start, when B is just behind A, when B is beside A, and when B pulls in; at each pause ask: 'What do you see right now?' and WAIT TIME of 10-15 seconds before taking answers. Cold-call 5 learners, one for each pause. Say: 'Both cars show a high speed. Yet from Wanjiru's seat, Mr Kamau's car looks slow. Does that bother anyone?' Do NOT give any measured values, split times or formulas.	Notice-and-Wonder with a pause-and-probe video protocol: separating what learners notice (observation) from what they wonder (questions) trains evidence-based description, and pausing at key moments focuses attention on changes in speed and position that the later mathematics will describe. Repeated viewing builds a shared experience for the whole class.	Look for: (a) observations that distinguish the steady car A from car B that speeds up; (b) mention of direction, for example the lorry coming the opposite way; (c) use of everyday words 'speed up' and 'slow down' correctly; (d) learners checking their earlier predictions against the clip. Listen for the key puzzle being voiced: 'Both cars are fast, but A looks slow from B.' Note learners who record opinions ('Wanjiru drives badly') rather than observations.

Explain Phase  VIDEO: Position vs. time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Calculating Acceleration from Velocity and Time (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of four, learners discuss the puzzle: how can two vehicles that are both moving fast appear to move slowly relative to each other? They also discuss how long the overtake took, how far the cars travelled, and what an oncoming lorry has to do with it. Each group uses everyday explanations and sketches (no formulas) to produce an Initial Explanation Poster with three parts: (1) what we think is happening to car A, car B and lorry L; (2) our best idea about why A looks slow from B; (3) what we do not yet know. Groups revise their original prediction if the clip changed their thinking, marking the change in a different colour. Groups do a short gallery walk to see other posters.	Say: 'Talk to your group. Why does Mr Kamau's car look slow from Wanjiru's seat, when its speedometer is high?' Give 5 minutes for discussion and 5 minutes for the poster. Circulate with the prompts: 'What is car B doing that car A is not?' and 'Which vehicle is coming towards you, and does that change how fast it looks?' Do NOT confirm or correct ideas; say: 'We will test this idea with evidence in later lessons.' Run a 4-minute gallery walk. After it, WAIT TIME of 10-15 seconds, then ask: 'What ideas did you see on more than one poster?' Cold-call 3 groups to share one idea and one unanswered question. Record shared ideas and disagreements on the board, such as 'B speeds up', 'A stays steady', 'the lorry makes it risky'.	Initial Models with Gallery Walk: asking learners to commit to an explanatory sketch before they have the tools to explain exposes partial ideas and misconceptions and creates a need to know. Comparing posters shows that different explanations compete, which motivates the investigations that follow in the unit.	Look for: (a) posters that include car A, car B and lorry L in correct relative positions and directions; (b) explanations that link speeding up of B to passing A; (c) any attempt to link relative position to time and distance; (d) honest 'what we do not know' statements such as 'how long it takes' or 'how far B travels'. Note misconceptions such as 'a faster speedometer means the car must look faster to everyone' or 'the lorry has no effect on the overtake'.
Driving Question Board (DQB) Creation  VIDEO: Intro to vectors and scalars <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Acceleration and velocity Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners individually write 2 or 3 questions on sticky notes about the overtake, in the form 'How...?', 'Why...?', 'How long...?', 'How far...?'. They place them around the Driving Question on the board: 'How can we use distance, time, speed, velocity and graphs to describe and predict how one car overtakes another on a Kenyan road?' In groups, they sort the notes into categories such as 'How far and how long', 'How fast', 'Looking from another car', 'Safety and the lorry', and 'Graphs and numbers'. Each group chooses one question it most wants to answer and explains why. Learners note that some questions link to the two KICD key inquiry questions: how to find velocity and how to find acceleration.	Place the Driving Question on a large chart and read it aloud. Say: 'What questions do you have that would help us answer this?' Give 4 minutes of silent writing, with the prompt: 'Think about what you would need to know to describe and predict the overtake.' Collect and group notes with learners' help, asking: 'Where does this one belong? Why?' Use WAIT TIME of 10-15 seconds after asking: 'Which questions look like they need numbers? Which need graphs?' Cold-call 5 learners to read a question aloud and place it. Highlight any questions about 'how far', 'how long', 'how fast' and 'is it safe', and tell learners: 'We will begin with these next lesson.' Do NOT answer questions now; say: 'We will return to this board in every lesson and move notes when we can answer them.' Keep	Driving Question Board: collecting and categorising learner-generated questions gives learners ownership of the learning path and a public record of what the class wants to figure out. Sorting questions into categories reveals which ideas (distance, time, speed, velocity, acceleration, graphs, relative motion) the class will need to build.	Look for: (a) questions that are specific and testable, for example 'How long does the overtake take?' rather than 'Why do cars go fast?'; (b) questions that mention measurable quantities such as time, distance or speed; (c) questions about relative view ('Why does A look slow from B?') and safety with the oncoming lorry; (d) learners placing notes in sensible categories and justifying them. Record the number of learners who contributed at least one question and note learners who need support with question-writing.

		the DQB visible in the room.		
Model Building Phase  VIDEO: Interpreting direction of motion from velocity-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Average velocity and average speed from graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners create an Initial Model Sketch on a half page of paper or in their notebooks. It shows a straight road near Naivasha, car A, car B, lorry L, arrows for direction of motion (Nakuru direction for A and B, opposite direction for L), and labelled positions at the start, in the middle and at the end of the overtake. Next to each vehicle they write whether it is speeding up, steady or slowing down, using everyday words. They add 'What I think will matter' with a list of ideas (for example time, distance, speed, gap to the lorry). They mark one thing they are least sure about with a star. They keep the sketch in their notebooks to revise in later lessons.	Say: 'Draw the first version of your model of the overtake. It is meant to change as we learn more.' Show a simple example layout without data: a straight line for the road and three small boxes for A, B and L. Use WAIT TIME of 10-15 seconds before asking: 'What should you include so that someone else could understand your model?' Cold-call 3 learners to name one feature to include. Circulate for 5 minutes and ask: 'How does your model show that B speeds up but A does not?' and 'Where is the lorry at the end of the overtake?' Close with: 'Keep this model safe. In Lesson 2 we start with your questions about how far and how fast.' Remind learners that overtaking on real roads is dangerous and that they should always keep safe on roads. Do NOT introduce formulas or numbers beyond 72 km/h.	Progressive Model Building: an initial sketch model records current understanding in words and pictures, which can be revised later with graphs and calculations. Marking uncertainty with a star links the model to the DQB and creates a purpose for upcoming lessons.	Look for: (a) models that show all three vehicles and the correct directions; (b) labels showing car B speeding up and car A steady; (c) a list of ideas that matter, with some that are measurable; (d) a starred area of uncertainty that connects to the DQB. Scan for learners whose sketch omits the lorry or shows both cars moving in opposite directions and follow up in Lesson 2.

D. TEACHER REFLECTION

After the lesson, consider: Did learners make predictions with numbers and reasons? Which prior ideas about speed, distance and time were most common? Did learners voice the puzzle that both speedometers are high while A looks slow from B? Which DQB questions came from learners rather than from you? Were any learners dominating while others stayed silent, and how did cold-calling help? Which questions about 'how far' and 'how fast' should open Lesson 2? Did learners stay safety-minded? Note any learners who need support with sketching or question-writing, and keep the predictions and sketches for comparison in Lesson 10.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe in the clips? Describe in your own words what car A (blue Toyota Axio), car B (white Subaru) and the oncoming lorry L were doing, and what happened when B returned to its lane.
What did I learn?	What did we notice that puzzled us? Record how the speedometers looked, how car A looked from car B, and what questions we

	now want to answer about how long, how far and how fast.
How does this explain the phenomenon?	What is our first explanation? Write your best current idea for why car A looks slow from car B even though both speedometers are high, and why the oncoming lorry matters, and say what you still need to find out.

LESSON 2 (80 minutes (2 x 40-minute lessons)): Distance, Displacement, Speed, Velocity and Acceleration: Describing Motion on Kenyan Roads

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners distinguish distance from displacement and speed from velocity, and describe acceleration as the rate of change of velocity (qualitatively, with units), using a boda boda trip, the Naivasha Express matatu, the cyclist Atieno and the overtaking on the Naivasha highway, so they can start answering the DQB questions about 'how far' and 'how fast'.
Knowledge	Distance is the total path length travelled (scalar, m or km). Displacement is the straight-line change in position with direction (vector, m or km). Speed = distance/time (scalar). Velocity = displacement/time (vector, with direction shown by sign; the positive direction is towards Nakuru). Velocity is uniform (constant) when speed and direction do not change and non-uniform when either changes. Acceleration is the rate of change of velocity with time, measured in m/s^2 . It is positive when an object speeds up, negative (deceleration) when it slows down, and zero when velocity is constant. Car A has constant velocity (20 m/s), car B accelerates (20 to 30 m/s), and the oncoming lorry L has constant velocity of -20 m/s.
Skills	Classifying quantities as scalar or vector; reading a motion story and finding distance and displacement; comparing speed and velocity; using sign to show direction; asking questions and defining problems; analysing and interpreting data (NGSS SEP); constructing explanations using evidence; communicating reasoning in pairs and whole class.
Attitudes	Curiosity about how vehicles move on Kenyan roads; precision in using scientific words; respect for others' reasoning during discussion; responsibility and safety awareness on the road; perseverance when a result looks surprising (displacement 0 after a 12 km ride).
Key Inquiry Question	1. How do you determine the velocity of a moving body? 2. How do you determine the acceleration of a moving body? (This lesson builds the meanings; calculations come in Lesson 3.)
Purpose in Storyline	Lesson 1 left learners with DQB questions about 'how far' and 'how fast' and puzzling speedometers. This lesson gives the vocabulary and the distinctions needed to describe motion precisely. It sets up Lesson 3 (calculating velocity and acceleration) and the graph lessons.
Safety Notes	Classroom walking activity: clear desks and bags from the walkway and walk at normal pace only. When discussing road scenes, stress that learners must never judge distances or overtake while cycling or riding. Use the real clip only for analysis, not to encourage risky behaviour.

B. LESSON OVERVIEW

Learners start by predicting how far a boda boda rider Juma is from the Naivasha stage after a trip and whether distance and displacement are the same. They then gather evidence by walking out-and-back paths in the classroom and by analysing three Kenyan journeys: Juma's boda boda trip (4 km distance, 2 km displacement east), the Naivasha Express matatu (90 km in 1.5 h) and the cyclist Atieno (12 km distance, displacement 0). In the Explain phase they sort quantities into scalars and vectors, define speed, velocity and acceleration, and classify cars A, B and lorry L from the overtaking clip as constant velocity or accelerating. They add new questions and answers to the Driving Question Board and revise the initial model of the overtake with the new vocabulary. No acceleration calculations are done in this lesson.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Intro to vectors and scalars Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Velocity and Acceleration (Flexbook) Source: CK-12  Search ARES for similar readings	(10 min) Learners recall the overtaking clip on the Nairobi-Nakuru highway near Naivasha and the DQB questions on 'how far' and 'how fast' from Lesson 1. They read the story of Juma the boda boda rider: he leaves the Naivasha stage, rides east to the market, then turns back west to the school. Individually they predict in their notebooks: (a) Is the total length he rode the same as how far he is from the stage at the end? (b) Is a speedometer reading the same as saying 'going 72 km/h towards Nakuru'? They draw a quick sketch of Juma's path and write a prediction with a reason. No numbers are given yet.	Display the DQB and read aloud two Lesson 1 questions about 'how far' and 'how fast'. Say: 'Yesterday we wondered how far and how fast. Today we will find out whether those words mean exactly what we think.' Read Juma's story slowly without numbers. Ask: 'Without calculating, is the length of the road Juma rode the same as how far he ends from the stage? Write your prediction and one reason.' WAIT TIME: 15 seconds of silent thinking before any talk. Then pair-share for 2 minutes. Cold-call 4 learners from different parts of the room and record predictions in two columns on the board ('Same' / 'Different'), without judging. Ask: 'Achieng in the back seat of car B reads 72 km/h on the speedometer. What does the speedometer not tell her?' WAIT TIME: 10 seconds. Record ideas about direction.	Eliciting prior ideas through Predict-Share (Predict-Observe-Explain). Writing a prediction with a reason before instruction exposes everyday meanings of 'distance' and 'speed' and creates a need for precise definitions. Recording both predictions publicly lets learners compare their ideas with evidence later.	Check notebooks for (1) a written prediction with a reason, (2) a sketch of the path showing the turn-back, (3) any mention of direction. Look for the common misconception that 'distance and displacement are always equal' or that 'speed tells everything about motion'. Tally predictions on the board (Same / Different) to compare with the evidence later.
Observe Phase  VIDEO: Calculating average velocity or speed Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Velocity and Acceleration Practice Source: CK-12  Search ARES for similar readings	(25 min) Station 1, Walk the Path (10 min): In groups of 4, learners mark a 'stage' line on the floor with chalk or tape and walk a path in the classroom: 3 m east, then 1 m back west (scaled-down version of Juma's trip, 1 km = 1 m). One learner walks, one measures the length of path walked (distance), one measures the straight line from start to end with direction (displacement), and one records. Station 2, Journey Data Cards (15 min): groups analyse three cards. Card 1: Juma leaves the Naivasha stage, goes 3 km east to the market in 10 min, then 1 km west to the school in 5 min more. Card 2: Naivasha Express matatu, Nairobi to Naivasha, road distance	Introduce stations: 'You are motion detectives. For each journey, find two different answers to the question: how far?' Circulate to Station 1 and ask: 'Show me with your hand the path you walked. Now show me the straight line from start to finish. Are they the same?' Use WAIT TIME of 10 seconds after each question. At Station 2 prompt: 'Atieno ends at her gate at home. Where is she compared with where she started? What does that say about her displacement?' If groups say '12 km', ask: 'Is that the path length or the change in position?' Pull groups together for 3 minutes: cold-call 3 groups to report their table for Juma, the matatu and	Collaborative data analysis with structured tables (Analysing and Interpreting Data, NGSS SEP 4) combined with an embodied activity. Physically walking out-and-back links the abstract ideas of path length and change in position to learners' own movement, and then the same structure is found in familiar Kenyan journeys. The three-column comparison (what it tells / does not tell) makes the distinction between scalar and vector explicit.	Observe whether groups: (1) correctly record Juma's distance as 4 km and displacement as 2 km east of the stage; (2) record Atieno's displacement as 0 while her distance is 12 km; (3) give the matatu's distance as 90 km and time 1.5 h without inventing direction numbers. Observable indicators: learners trace the path with a finger and then draw a straight arrow from start to end. Note groups who confuse 'displacement' with 'total length' and prompt them with the Station 1 walk.

	90 km in 1.5 h. Card 3: Atieno cycles from home to the Lake Naivasha gate 6 km away and back home in 1 hour (12 km of cycling). For each card, learners fill a table with columns: total path length (distance), final position relative to start (displacement and direction), total time, and 'what does the number tell us / not tell us?' Expected: Juma distance 4 km, displacement 2 km east of the stage; matatu distance 90 km, 1.5 h; Atieno distance 12 km, displacement 0 km. Learners also note that the matatu's 90 km is the road length and mention that its direction is towards Naivasha (north-west from Nairobi).	Atieno. Ask: 'Why can Atieno cycle 12 km and still have displacement 0?' WAIT TIME: 15 seconds. Revisit the Predict board: ask 2 learners to move any prediction they now want to change. Do NOT yet introduce formulas for acceleration.		
Explain Phase  VIDEO: Intro to vectors and scalars Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Calculating Acceleration from Velocity and Time (Flexbook) Source: CK-12  Search ARES for similar readings	(25 min) Learners use their data to build meanings. Step 1 (8 min): Using the three journeys, groups complete sentence frames: 'Distance is ... It has/does not have direction. Displacement is ... It has/does not have direction.' Step 2 (7 min): Speed and velocity. Using Juma's trip, learners discuss: speed uses distance, velocity uses displacement and direction. They note that the matatu's average speed is found from 90 km and 1.5 h (to be calculated formally in Lesson 3), and that Atieno's average velocity must be 0 because her displacement is 0, even though she was cycling. They agree the positive direction is towards Nakuru, so lorry L, heading to Naivasha, has a negative velocity. Step 3 (10 min): Acceleration and the overtaking clip. Learners get a card describing cars A, B and lorry L: A steady on the speedometer at 72 km/h, B speeding up from 72 km/h to 108 km/h, L steady at 72	Say: 'We have three words that sound alike: distance, speed, and velocity. Let us decide what makes them different.' Run a Think-Pair-Share on each step. Step 1: ask 5 cold-called learners to read their completed frames; record the class definitions on the board in two columns: Scalar (distance, speed) and Vector (displacement, velocity). Ask: 'What is missing from speed that velocity has?' WAIT TIME: 10 seconds. Step 2: ask: 'If Achieng's speedometer says 72 km/h in car B, and lorry L also shows 72 km/h, do they have the same velocity? Why or why not?' WAIT TIME: 15 seconds, then cold-call 3 learners. Correct gently: lorry L velocity is -20 m/s because it travels towards Naivasha. Remind them 72 km/h = 20 m/s (quote the conversion only as a fact, calculation practice comes in Lesson 3). Step 3: show the A, B, L card. Ask: 'Whose velocity changes during the overtake?' WAIT TIME: 10	Claim-Evidence-Reasoning and Sorting/Classifying (Constructing Explanations, NGSS SEP 6). Learners use the evidence from the journey cards to claim whether each quantity is scalar or vector and justify it with reasoning about direction. Classifying A, B and L ties the new vocabulary directly to the anchoring phenomenon, building the language needed to explain why B's speedometer changes while A's does not.	Check the sentence frames for: distance = total path length, no direction; displacement = straight-line change in position with direction. Listen for correct classification: A constant velocity, L constant velocity (negative), B accelerating (positive). Indicators of understanding: learner says Atieno's average velocity is 0 although her speed was not 0; learner uses 'negative' for the lorry because of direction, not because it is slow. Exit prompt on a slip: 'Give one example from the Naivasha highway of a scalar and a vector.' Look for misconceptions such as 'negative velocity means slowing down' and address them in the next phase.

	km/h towards Naivasha. They classify each: constant velocity or accelerating? Which has positive acceleration? What happens to the velocity of B? They draw a 'velocity story' for each vehicle using arrows (length = size of velocity, direction = sign) and write one sentence defining acceleration as the rate of change of velocity with time, with unit m/s^2, and the meaning of positive, zero and negative acceleration. They also ask: does lorry L have acceleration? (No, its velocity does not change, but its velocity is negative.) They state that a vehicle driving round a bend at a steady speed would still be accelerating because its direction changes (non-uniform velocity), but that on this straight road that does not occur. No numerical acceleration calculations are done.	seconds. 'Who has acceleration? Who has zero?' Cold-call 4 learners. Introduce: 'Acceleration is how quickly velocity changes. Its unit is m/s^2.' Explain deceleration is negative acceleration. Say: 'We will calculate acceleration next lesson; today we only decide when it is positive, zero or negative.' If asked for numbers, reply: 'Hold that question for Lesson 3; write it on the DQB.'		
Driving Question Board (DQB) Creation  VIDEO: Calculating average velocity or speed Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Calculating Acceleration from Velocity and Time Practice Source: CK-12  Search ARES for similar readings	(10 min) Learners return to the DQB from Lesson 1. In pairs they sort the existing questions on 'how far' and 'how fast' into three groups: those they can now partly answer (e.g. 'Is distance the same as how far the car ends up?'), those that need calculations (e.g. 'How fast is B going compared with A?'), and those that need new tools such as graphs or timing. Each pair writes one new question on a sticky note that arose today, for example: 'How do we calculate acceleration of B?', 'How can we show Atieno's trip on a graph?', 'How long does the overtake take and how far do the cars go?' They place notes on the DQB and add a green dot to questions that now have a partial answer.	Point to the DQB: 'Which of our questions did today's evidence help us answer, even a little?' WAIT TIME: 10 seconds. Cold-call 3 pairs to move a question and explain why with evidence from Juma, the matatu or Atieno. Ask: 'What new question do you have about car B speeding up from 72 km/h?' Collect at least 6 new sticky notes. Link questions to future lessons: 'Calculating velocity and acceleration is coming in Lesson 3; graphs after that.' Ensure two KICD questions are visible on the board: 'How do you determine the velocity of a moving body?' and 'How do you determine the acceleration of a moving body?' and mark them as 'in progress: we know what they mean, not yet how to calculate'.	Driving Question Board as a public record of growing understanding. Sorting questions by what is answered, what needs calculation and what needs graphs shows learners that science builds over several lessons and keeps the investigation purposeful and student-driven.	Count how many pairs correctly place at least one question into 'partly answered' with a reason from today's evidence. Check new questions for quality: are they about measuring, calculating or predicting the overtake? Look for questions that show confusion about direction or acceleration, and plan to address them in Lesson 3.

Model Building Phase  VIDEO: Position vs. time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Acceleration and velocity Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(10 min) Learners revise their initial model of the overtake from Lesson 1 (a labelled sketch of cars A, B and lorry L on the Naivasha road). They add: (1) an arrow for the positive direction towards Nakuru; (2) velocity arrows for A (constant, 72 km/h), B (getting longer as it speeds up from 72 km/h to 108 km/h), and L (pointing the opposite way, labelled negative); (3) a short note naming which vehicle has positive acceleration and which have zero acceleration; (4) a note that distance travelled by B is not the same thing as how far ahead of A it ends up. They write a two-sentence explanation: 'Car B is accelerating because ... Car A has constant velocity because ...'. They leave space for numbers to be added in later lessons.	Say: 'Take out your Lesson 1 model. Use today's vocabulary to make it more precise.' Model one example on the board: draw arrows for A and L and ask: 'Which way should L's arrow point and why?' WAIT TIME: 10 seconds. Circulate and ask each group: 'How does your model show that B is accelerating and A is not?' Cold-call 3 learners to share their explanation sentence. Ask: 'What can your model not tell us yet?' Expect: how long it takes, how far, the exact acceleration, and whether it is safe. Record these on the board as 'next steps'. Collect models for the final model in Lesson 10.	Developing and Using Models (NGSS SEP 2) with iterative revision. Adding velocity arrows and labelled directions to the Lesson 1 sketch turns everyday observations into a scientific representation and makes gaps in understanding visible, motivating the calculations and graphs in later lessons.	Check models for: (1) a direction convention with Nakuru as positive; (2) lorry L labelled with a negative velocity; (3) B shown with changing velocity and A with constant velocity; (4) correct use of the words distance, displacement, speed, velocity and acceleration. Use a 3-point check: all four present (secure), two or three (developing), fewer (needs support).
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D. TEACHER REFLECTION

Which learners still used 'distance' and 'displacement' as the same word after the Atieno example, and which example convinced them? Did learners separate 'negative velocity' (direction) from 'slowing down' (negative acceleration), especially for lorry L? Which DQB questions arose that signal readiness for calculations in Lesson 3, and which learners need more practice with the positive direction towards Nakuru? How effectively did the walking activity connect to the Kenyan journeys, and how could the time allocation be adjusted to give more discussion time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Juma's boda boda trip: 4 km distance but 2 km displacement east of the stage. Naivasha Express matatu: 90 km in 1.5 h. Cyclist Atieno: 12 km distance but displacement 0. In the overtaking clip, car A stays steady at 72 km/h, car B speeds up from 72 km/h to 108 km/h, and oncoming lorry L travels the opposite way.
What did I learn?	Distance is total path length (scalar); displacement is straight-line change in position with direction (vector). Speed uses distance; velocity uses displacement and direction. Acceleration is the rate of change of velocity in m/s^2 : positive when speeding up, zero for constant velocity, negative (deceleration) when slowing down.
How does this explain the phenomenon?	Car A has constant velocity so zero acceleration. Car B has positive acceleration because its velocity increases. Lorry L has constant negative velocity because it travels towards Naivasha, opposite to the positive direction towards Nakuru, and zero acceleration. A speedometer shows speed only, so velocity is needed to describe direction as well.

LESSON 3 (80 minutes (2 x 40-minute periods)): Calculating Velocity and Acceleration: How Fast, and How Quickly Does the Speed Change?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners calculate velocity using velocity = displacement/time and acceleration using $a = (v - u)/t$, converting km/h to m/s and using the negative sign for deceleration and for motion in the opposite direction, so that they can put real numbers on the overtaking car B, the overtaken car A and the oncoming lorry L.
Knowledge	Velocity = displacement / time (vector; sign shows direction, positive towards Nakuru). Acceleration $a = (v - u)/t$ in m/s^2 . Deceleration is negative acceleration. Convert km/h to m/s by dividing by 3.6 and m/s to km/h by multiplying by 3.6 (72 km/h = 20 m/s, 108 km/h = 30 m/s). Zero acceleration means constant velocity.
Skills	Using mathematics and computational thinking (NGSS SEP 5): substituting into formulas, converting units and checking signs. Analysing and interpreting data (SEP 4): reading values from the clip data. Constructing explanations (SEP 6): explaining why A has zero acceleration and L has negative velocity. Communication and collaboration through pair and whole-class talk.
Attitudes	Learners show precision and care with units and signs, persevere with multi-step calculations, and appreciate that mathematics helps us judge real road situations such as overtaking safely.
Key Inquiry Question	1. How do you determine the velocity of a moving body? 2. How do you determine the acceleration of a moving body?
Purpose in Storyline	Lesson 2 gave learners the definitions of distance, displacement, speed, velocity and acceleration (qualitatively, with units). Lesson 3 turns those definitions into calculations using the anchor data: A 150 m in 7.5 s gives 20 m/s, B accelerates from 20 to 30 m/s in 5 s giving 2 m/s^2 , and the lorry has velocity -20 m/s. These numbers are the evidence that later lessons use for graphs (Lessons 4-7), relative speed (Lesson 8) and the safety decision (Lesson 9).
Safety Notes	Do not take learners to a roadside to measure real vehicles in this lesson. Use the video clip and data cards only. Keep the classroom aisles clear during pair-movement activities. If a short walking or trolley-timing demonstration is used, do it in the school compound away from any road.

B. LESSON OVERVIEW

Learners begin by predicting the numerical velocity of Car A and the acceleration of Car B from the Naivasha overtaking clip, then work through the anchor data to calculate A's velocity (150 m in 7.5 s = 20 m/s), convert speedometer readings (72 km/h = 20 m/s and 108 km/h = 30 m/s), calculate B's acceleration $((30 - 20)/5 = 2 \text{ m/s}^2)$ and assign the lorry a velocity of -20 m/s because it travels towards Naivasha. They then apply the same methods to two supporting situations, the boda boda first leg (18 km/h) and the braking matatu ($a = -2 \text{ m/s}^2$), before adding the new evidence to the DQB and the class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the Naivasha	Display the still frames of A, B and	Predict-Observe-Explain with a	Check the prediction cards and

 VIDEO: Calculating average velocity or speed Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Calculating Acceleration from Velocity and Time (Flexbook) Source: CK-12  Search ARES for similar readings	overtaking clip (Wanjiru in white Subaru B overtaking Mr Kamau's blue Toyota Axio A, with Mr Otieno's lorry L coming towards them). On a prediction card they individually write: (a) the speed they think A is travelling, in m/s, if its speedometer reads 72 km/h; (b) whether B's speed changes by the same amount every second; (c) what sign (+ or -) they would give the lorry's velocity and why, if positive is towards Nakuru. They share predictions with a partner and place a coloured sticker on a class tally chart (e.g. A: 20 m/s, 72 m/s, 7.2 m/s, don't know).	L with arrows showing direction (positive towards Nakuru). Say: 'Yesterday we separated distance from displacement and speed from velocity. Today we put numbers on them. Before I show you any numbers, write your best prediction. There are no wrong predictions yet.' Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds after asking: 'Why might the lorry's velocity need a different sign from A's?' Cold-call 4 learners (not volunteers) to share, and ask 2 of them: 'What made you choose that?' Record all ideas on the board without judging them. Close with: 'Keep your card. We will test these predictions with evidence from the clip.'	tally chart. Committing to a written prediction exposes misconceptions (for example that 72 km/h equals 72 m/s, or that the lorry's velocity is positive because it is fast), so learners have a reason to look closely at the evidence and can later compare their old ideas with their new ones.	tally chart. Look for: learners who write a number with a unit for A; learners who mention direction when discussing the lorry; learners who treat speed and velocity as different ideas (from Lesson 2). Note how many believe km/h and m/s are interchangeable, and how many think B's speed jumps suddenly rather than changing steadily. Use this to decide how long to spend on unit conversion.
Observe Phase  VIDEO: Acceleration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Calculating Acceleration from Velocity and Time Practice Source: CK-12  Search ARES for similar readings	In groups of 4, learners receive a data card from the clip (Car A: starts at the start point and travels 150 m in 7.5 s; Car B: speedometer 72 km/h at $t = 0$ and 108 km/h at $t = 5$ s; Lorry L: travels 150 m towards Naivasha in 7.5 s). Learners replay the clip segments or study the frames. Station 1: calculate A's velocity: $150 \text{ m} / 7.5 \text{ s} = 20 \text{ m/s}$ towards Nakuru (+20 m/s); check this against the speedometer 72 km/h by dividing 72 by 3.6 = 20 m/s. Station 2: convert B's speedometer readings: $72 \text{ km/h} / 3.6 = 20 \text{ m/s}$; $108 \text{ km/h} / 3.6 = 30 \text{ m/s}$; then calculate $a = (v - u)/t = (30 - 20)/5 = 2 \text{ m/s}^2$. Station 3: calculate the lorry's velocity: displacement 150 m towards Naivasha = -150 m, so $v = -150/7.5 = -20 \text{ m/s}$ (speed 20 m/s, velocity -20 m/s). Groups record the working in a table: quantity, formula, substitution, answer with unit and sign. Learners also use the split-time table to check B's	Circulate with a checklist. Ask: 'What is the displacement, and in which direction?' and 'Which unit must be in metres per second before you substitute?' Insist on units in every line of working. When a group gets 72 m/s for A, say: 'Does 72 m/s match the distance of 150 m in 7.5 s? Check it.' Do not give the answer; let the clip data correct them. At each station, WAIT TIME of 10-15 seconds after posing: 'Why does the lorry get a negative sign when its speed is the same as A's?' Cold-call 3 groups to report their station results. Ask one group to explain why A's acceleration is $(20 - 20)/7.5 = 0 \text{ m/s}^2$. Prompt: 'Does zero acceleration mean zero velocity?'	Structured data analysis with a worked-calculation table (Using Mathematics, SEP 5). Using the same layout for each body (formula, substitution, answer with unit and sign) makes the structure of the method visible, and comparing A, B and L side by side builds the idea that velocity has direction and that acceleration measures change in velocity, not velocity itself.	Observe whether learners: (1) divide by 3.6 to convert km/h to m/s and can say why; (2) write units on every line; (3) give the lorry -20 m/s and not +20 m/s; (4) calculate B's acceleration as 2 m/s^2 using $(v - u)/t$, not as $30/5 = 6$ or $(30 + 20)/5$; (5) recognise that A has zero acceleration yet a non-zero velocity. Record a tick or note for each group on the checklist.

	velocity rises by 2 m/s every second (20, 22, 24, 26, 28, 30) and that A's velocity stays 20 m/s at every time (zero acceleration).			
Explain Phase  VIDEO: Calculating average velocity or speed Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Velocity and Acceleration (Flexbook) Source: CK-12  Search ARES for similar readings	Learners apply the methods to two new Kenyan situations and then explain their results in terms of the overtaking. Task 1 (boda boda Juma, Naivasha town): on the first leg Juma rides 3 km east in 10 min. Learners calculate velocity in km/min (0.3 km/min), convert to km/h (18 km/h east, +18 km/h) and then to m/s ($18 / 3.6 = 5$ m/s). Task 2 (braking matatu): the matatu slows from 20 m/s to 0 in 10 s. Learners calculate $a = (0 - 20)/10 = -2 \text{ m/s}^2$ and write 'deceleration 2 m/s^2 '. Learners then write a 3-4 sentence explanation, using the words velocity, acceleration, direction and sign, answering: 'How do the numbers 20 m/s, 2 m/s^2 and -20 m/s describe A, B and L in the overtaking clip?' They also answer: 'What would B's acceleration be if Wanjiru had taken 10 s instead of 5 s to reach 30 m/s?' ($(30 - 20)/10 = 1 \text{ m/s}^2$, a gentler speed-up). Pairs swap explanations and give one piece of feedback.	Say: 'Juma and the matatu use the same two formulas as the cars on the highway. Show me.' Pose: 'When the matatu slows down, why is its acceleration negative if its speed is still positive?' WAIT TIME of 10-15 seconds, then cold-call 4 learners. Press on reasoning: 'Say that again using the words change in velocity.' Address the misconception that negative acceleration means moving backwards: sketch the matatu moving towards Nakuru with $a = -2 \text{ m/s}^2$. Ask: 'In what way is the lorry's negative velocity different from the matatu's negative acceleration?' Model one sentence stem: 'The velocity of ____ is ____ because ____.' Select two explanations to read aloud and ask the class to agree or add.	Claim-Evidence-Reasoning with pair peer feedback (Constructing Explanations, SEP 6). Learners must support each claim about a vehicle with a calculation and a reason about direction or change in velocity. Transferring the method to the boda boda and matatu shows that the formulas work for any straight-line motion, and contrasting negative velocity (direction) with negative acceleration (slowing down) addresses a common confusion.	Collect the written explanations. Success indicators: correct values ($18 \text{ km/h} = 5 \text{ m/s}$; $a = -2 \text{ m/s}^2$; 20, 2, -20); the word 'deceleration' used correctly; direction linked to sign; B's acceleration described as the change in velocity per second. Flag learners who write acceleration = 30 m/s or who say the lorry is 'slower' because its velocity is negative. Give an exit ticket: 'Convert 54 km/h to m/s and find the acceleration of a car going from 0 to 15 m/s in 5 s.' (15 m/s; 3 m/s^2 .)
Driving Question Board (DQB) Creation  VIDEO: Acceleration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Velocity and Acceleration	Learners return to the class DQB from Lesson 1 (questions about 'how far' and 'how fast'). In groups they sort the questions into: answered today (e.g. 'How fast is A really going?' answered: 20 m/s; 'Is B's speed changing steadily?' answered: yes, $a = 2 \text{ m/s}^2$), partly answered, and still open (e.g. 'How far do the cars travel?', 'Why does A look slow from B?', 'Is it safe with the lorry coming?'). They add new questions on sticky notes such as 'Can a graph show the	Say: 'Look at our board. Which questions did today's evidence help with? Which ones need new evidence?' Hand each group 2 coloured stickies: green for answered, yellow for new. WAIT TIME of 10-15 seconds after asking: 'Which question do you think we should answer next, and why?' Cold-call 3 groups to place a note and justify it by pointing to evidence. Highlight that the next lessons will use graphs (Lessons 4-7) to show these same numbers,	Driving Question Board sorting. Sorting questions by whether evidence has answered them makes learning visible, lets learners see progress towards the driving question and gives them ownership of what to investigate next.	Check that sorted questions are placed accurately (for example 'What is the velocity of A?' is marked answered, 'How long does overtaking take?' is not yet). Note whether new questions refer to graphs, distance covered or relative viewpoint, showing that learners are connecting to the next lessons. Look for learners who can link a question to specific evidence from today.

Practice Source: CK-12 Search ARES for similar readings	acceleration?' and 'What does A look like from B's seat?', and link each to the driving question about describing and predicting the overtaking.	and that relative speed will come later. Do not answer the open questions; add them to the 'Coming up' column.		
Model Building Phase  VIDEO: Position vs. time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Acceleration and velocity Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners add to their personal overtaking model (a labelled road diagram of A, B and L from earlier lessons). For each vehicle they add a velocity arrow with a sign and label: A: +20 m/s (72 km/h), zero acceleration; B: +20 m/s rising to +30 m/s (108 km/h), $a = 2 \text{ m/s}^2$ for 5 s, then 30 m/s steady; L: -20 m/s (72 km/h towards Naivasha). They also write the two formulas, velocity = displacement/time and $a = (v - u)/t$, in a 'Tools box' beside the model along with the conversion rule 'km/h / 3.6 = m/s'. Finally, learners complete the sentence: 'Our model now explains ____ but cannot yet explain ____.'	Say: 'Your model should now have numbers and directions, not just pictures. Add the new evidence and mark what you have changed since Lesson 2.' Circulate and ask: 'Where does the sign of the lorry show in your model?' and 'Where on your model does B's acceleration begin and end?' WAIT TIME of 10-15 seconds, then cold-call 3 learners to read their gaps sentence. Record common gaps (how far each car travels, how long the overtake lasts, why A looks slow from B) on the board, saying: 'These are exactly what the next lessons will answer.' Collect the models for review.	Cumulative model revision with a 'what it explains / what it cannot yet explain' statement. Updating the model with quantities makes learners connect each number to a physical feature of the overtaking, and naming its limits keeps the investigation going.	Review models for: correct values and signs (+20, +20 to +30, -20 m/s); acceleration 2 m/s^2 marked only on B during the speeding-up; zero acceleration for A and L; formulas and conversion rule stated correctly. Gaps statements should refer to distance, time or relative viewpoint. Provide feedback for learners who give B a constant 30 m/s throughout or mark the lorry as positive.

D. TEACHER REFLECTION

After the lesson, consider: Which learners confused speed with velocity or treated negative acceleration as moving backwards? Did learners use the 3.6 conversion with understanding, or only by memory? Did the clip data (150 m in 7.5 s giving 20 m/s matching 72 km/h) help learners correct their own predictions? Which cold-called learners struggled to explain the sign of the lorry's velocity and need a follow-up? What changes to pacing or grouping are needed before Lesson 4, when learners start drawing graphs of these quantities?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that Car A covers 150 m in 7.5 s, Car B's speedometer rises from 72 km/h to 108 km/h in 5 s and that the oncoming lorry L moves towards Naivasha. We observed that the matatu slows from 20 m/s to 0 in 10 s.
What did I learn?	We learned that velocity = displacement/time and $a = (v - u)/t$, that km/h is converted to m/s by dividing by 3.6, and that a negative sign shows direction (the lorry, -20 m/s) or deceleration (the braking matatu, $a = -2 \text{ m/s}^2$).
How does this explain the	We can explain that A has velocity +20 m/s with zero acceleration, B has acceleration 2 m/s^2 while it speeds up from 20 to 30

phenomenon?	m/s, and L has velocity -20 m/s because it travels in the opposite direction, so the numbers describe how each vehicle moves in the overtaking.
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LESSON 4 (80 minutes (2 x 40 minutes)): Drawing Displacement-Time Graphs: Parked, Steady and Returning Movements

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners choose suitable scales, plot points and draw straight-line displacement-time graphs for a stationary vehicle, a steady car and a returning cyclist, and describe the shape of each graph. This gives them the graph-drawing skill they will later use to describe the overtaking on the Naivasha highway.
Knowledge	Learners know that a displacement-time graph has time t on the x-axis and displacement on the y-axis. They know that a parked vehicle gives a horizontal line, a steady car gives a straight sloping line, and a returning movement rises, stays flat and then falls back to zero. They recall from Lessons 2-3 that displacement is a vector with direction and that velocity = displacement/time.
Skills	Choosing scales to fit a data table onto graph paper; labelling axes with quantity and unit; plotting points accurately; joining points with a ruler; describing the shape of a graph in words (Science and Engineering Practice: developing and using models; analysing data).
Attitudes	Learners show accuracy, neatness and perseverance when drawing graphs, and value careful checking and peer feedback.
Key Inquiry Question	How do you determine the velocity of a moving body? (This lesson prepares the graphs from which velocity will be read in Lesson 5.)
Purpose in Storyline	Lessons 2-3 gave learners the ideas of displacement and velocity = displacement/time. Lesson 4 turns movement into pictures. Learners draw three displacement-time graphs (parked matatu, steady car, returning cyclist Atieno) so that in Lesson 5 they can interpret them. The graph shapes are also a first step toward describing Car A's steady motion and Car B's changing motion in the overtaking clip.
Safety Notes	Use sharp pencils and rulers safely. Learners remain seated while plotting. Keep the walking-out activity in a clear corridor or playground, with no running.

B. LESSON OVERVIEW

Learners begin by predicting what a picture of a parked matatu, a steady car and a returning cyclist would look like on a graph of displacement against time. They then observe the data tables for three Naivasha situations: a parked matatu at a stage 5 km from Naivasha town for 30 min; a steady car at 60 km/h with points (0,0), (30,30), (60,60) in (min, km); and cyclist Atieno going to the Lake Naivasha gate and back with points (0,0), (30,6), (40,6), (60,0). In the Explain phase they choose scales, plot and draw the graphs and describe each shape using only words such as flat, rising and falling. They add questions to the Driving Question Board and revise the class model by adding 'a graph can show how displacement changes with time'. No gradient or velocity is read from the graphs in this lesson; that is Lesson 5.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Position vs. time graphs Source: Khan Academy (English)	Learners recall the overtaking clip (Car A, the blue Toyota Axio, steady at 72 km/h; Car B, the white Subaru, speeding up). On a blank axis sketch in their notebooks (t along the bottom,	Draw three empty axes on the board. Say: 'Kiprono films the road and we want a picture of each movement. Sketch a line for each situation. There is no wrong answer yet, I want your best	Predict-Observe-Explain (initial ideas elicitation): committing to a sketch before seeing data surfaces misconceptions (for example, that a parked vehicle gives a line going up, or that a graph is a picture of	Look for: (a) whether learners label t on the x-axis and displacement on the y-axis; (b) whether the parked matatu is drawn flat or sloping; (c) whether the cyclist's return is drawn falling. Tally on a

- US curriculum) Search ARES for similar videos HTML: Position-Time Graphs (Flexbook) Source: CK-12 Search ARES for similar readings	displacement up the side) they predict what the line would look like for three situations: (1) a matatu parked at a stage 5 km from Naivasha town for 30 min; (2) a car moving steadily at 60 km/h from Naivasha; (3) cyclist Atieno cycling from home to the Lake Naivasha gate and back home. They compare predictions with a partner and note one reason for each choice.	thinking.' Read each situation slowly. After the sketching, give WAIT TIME of 10-15 seconds, then cold-call 4 learners with different sketches for the cyclist. Ask: 'What made you draw it going down at the end?' Record two or three sketches on the board without judging them. Say: 'We will test these with data.'	the road). Learners later compare the sketch with their drawn graph, which builds understanding through conflict and confirmation.	class list how many sketch the parked matatu as a rising line, to target in the Explain phase.
Observe Phase VIDEO: Position vs. time graphs Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Position-Time Graphs Practice Source: CK-12 Search ARES for similar readings	Learners work in groups of 4 with a data card for each situation. Parked matatu: displacement stays 5 km from Naivasha town for 30 min, with points (0,5), (15,5), (30,5) in (min, km). Steady car at 60 km/h: (0,0), (30,30), (60,60). Cyclist Atieno: (0,0), (30,6), (40,6), (60,0). Each group also acts out the cyclist's trip: one learner walks slowly away along a corridor for 'half the time', stands still for a short time, then walks back to the start while a timekeeper calls the phases and a recorder notes the position at each call. Learners notice that the position goes up, stays the same, then goes back to zero.	Hand out the data cards and graph paper. Say: 'Look at the numbers first. Where does the cyclist's displacement go up, where does it stay the same, and where does it come back to zero?' Circulate and ask: 'Atieno ends at displacement 0. Where is she?' (back at home). Give WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to report what they notice in the table of the walking activity. Say: 'Positive means away from home; the return trip brings the displacement back to zero.' Remind learners that displacement has direction, as met in Lesson 2.	Embodied modelling with data tables: acting out the cyclist's trip connects the abstract numbers to a real movement, helping learners see why the displacement values rise, stay and fall before they plot them.	Listen for statements such as 'the displacement does not change from 30 to 40 minutes' and 'it returns to 0 at 60 minutes'. Check that each group correctly identifies the three phases of Atieno's journey in the table. Note groups that confuse distance travelled with displacement.
Explain Phase VIDEO: Interpreting direction of motion from position-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Step 1 (scales): Learners decide how to fit the data on graph paper. For the cyclist: x-axis 0 to 60 min, with 1 cm = 10 min; y-axis 0 to 6 km, with 1 cm = 1 km. For the steady car: y-axis 0 to 60 km, with 1 cm = 10 km. For the parked matatu: y-axis from 0 to 6 km, 1 cm = 1 km. Step 2 (plot and draw): They label both axes with quantity and unit, plot each point with a small cross, and join points with a ruler. Step 3 (describe): They write one sentence per graph: the parked matatu gives a horizontal	Model one scale choice on the board: 'The biggest time is 60 min and the page is 20 squares wide, so 1 cm for 10 min fits.' Ask: 'Why would 1 cm for 1 min not fit?' Give WAIT TIME of 10-15 seconds. Walk around with a checklist: axes labelled with units, scale even, points plotted exactly, ruler used. Cold-call 3 learners to describe a graph using only the words 'flat', 'rising' and 'falling'. Say: 'Today we describe the shape only. In the next lesson we will find out what the steepness tells us.' Do not	Scaffolded procedure with talk moves (revoicing and peer checking): learners follow the sequence choose scale, plot, join, describe, then compare drawn graphs with their Predict sketches. Revoicing ('So you are saying the line is flat because the matatu is not moving away') links vocabulary to meaning.	Observable indicators: scales are even and fit the page; axes have labels with units; all four cyclist points are correctly placed at (0,0), (30,6), (40,6), (60,0); the parked matatu line is horizontal; the steady car line passes through (30,30) and (60,60); descriptions use 'flat', 'rising' and 'falling' correctly. Collect one graph per learner as an exit check.

 HTML: Position-Time Graphs (Flexbook) Source: CK-12  Search ARES for similar readings	line at 5 km; the steady car gives a straight line rising from (0,0) to (60,60); the returning cyclist rises from (0,0) to (30,6), is flat to (40,6), then falls to (60,0). Step 4 (link): In pairs they discuss: 'The cyclist's line ends on the x-axis. What does that tell us about where Atieno is?' and 'How could Car A's steady motion in the overtaking clip look on a displacement-time graph?' (a straight rising line; shape only).	introduce gradient calculations. Ask: 'Which situation gives a line that comes back down? What does that tell us about the direction of the cyclist's movement?'		
Driving Question Board (DQB) Creation  VIDEO: Proportional relationships: graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Position-Time Graphs Practice Source: CK-12  Search ARES for similar readings	Learners revisit the Driving Question Board. They place the new graphs beside the overtaking clip and add or answer questions, for example: 'What would Car B's graph look like when it speeds up?', 'How can we tell from a graph how fast the cyclist went?', 'Can we use graphs to know when two vehicles meet?' They mark questions about graph shape as partly answered, and move questions about steepness and meeting points to a 'coming next' column.	Say: 'Look at the board. Which of our questions can the graphs help us answer today?' Give WAIT TIME of 10-15 seconds and cold-call 3 learners to read or add a question. Ask: 'Which questions need us to read the graph, not just draw it?' Place sticky notes under the labels 'answered in part' and 'next lesson'. Say: 'Next lesson we will read the cyclist's graph to find velocity.'	Driving Question Board tracking: showing what has been learned and what remains builds ownership and keeps the phenomenon (how one car overtakes another) connected to each skill. Sorting questions by status makes progress visible.	Check that new questions refer to the graph skills (shape, steepness, meeting, speeding up) and to the overtaking. Note learners whose questions show they see a graph as a tool for describing and predicting motion.
Model Building Phase  VIDEO: Points on the coordinate plane examples <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)	Learners update their class model for the driving question. In their notebooks they add a section titled 'Graphs of motion' with the three graphs drawn small, each labelled with its situation and shape: parked matatu (flat line), steady car (straight rising line), returning cyclist (rises, flat, falls to zero). Under them they write: 'A displacement-time graph shows how displacement changes as	Say: 'Add what you can now show that you could not show before. Use the words flat, rising and falling.' Walk round and ask: 'Which shape would Car A have?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to share their model additions. Ask: 'What is still missing from our model of the overtaking?' Expect: speeds, steepness, Car B's change, relative motion. Close	Cumulative model revision: adding the new graph tool to the model, and naming what is still missing, helps learners see how each lesson contributes to the final answer to the driving question.	Check each notebook for: three correctly labelled graph sketches; accurate shape descriptions; a clear statement of what a displacement-time graph shows; an identification of at least one gap in the model. Use a 3-point rubric (complete, partly complete, needs support).

Search ARES for similar videos EXERCISE: Graph labels and scales <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	time passes.' They add a sketch showing where Car A, which moves steadily, could be placed on such a graph, and a note: 'Car B's motion changes; we still need a way to show this.' Then they complete the summary table prompts.	with: 'Next lesson we interpret these graphs.'		
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D. TEACHER REFLECTION
Which learners chose workable scales without help, and which needed the 'biggest value fits the page' prompt? Did the walking activity help learners see why the cyclist's line returns to zero? Which learners treated the graph as a picture of the road? Were you able to keep the focus on drawing and describing shape, without drifting into gradient calculations that belong to Lesson 5? What would you change in the data cards to support learners who plot slowly?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed data for a parked matatu 5 km from Naivasha town, a steady car at 60 km/h with points (0,0), (30,30), (60,60), and cyclist Atieno with points (0,0), (30,6), (40,6), (60,0) in (min, km). We saw that the displacement stays the same, rises steadily, or returns to zero.
What did I learn?	We learned to choose scales, label axes with units, plot points and join them with a ruler to draw displacement-time graphs. A parked vehicle gives a horizontal line, a steady car gives a straight rising line, and a returning cyclist gives a line that rises, goes flat and then falls to zero.
How does this explain the phenomenon?	We can draw a graph of displacement against time for a stationary, steady or returning movement and describe its shape. This helps us describe the steady motion of Car A in the overtaking, and shows we need more tools, such as reading steepness and velocity-time graphs, to describe Car B speeding up.

LESSON 5 (40 minutes (1 lesson)): Interpreting Displacement-Time Graphs: Gradient, Rest and Meeting Points

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners read information from displacement-time graphs: velocity as the gradient (sign shows direction), rest as a flat line, and the meeting point of two bodies as the intersection of their graphs. They use this to move closer to describing and predicting the overtake on the Naivasha highway.
Knowledge	Learners will be able to: (1) state that the gradient of a displacement-time graph is velocity; (2) explain that a positive gradient means moving towards Nakuru (or away from home) and a negative gradient means moving the other way; (3) identify a flat line as rest (velocity 0); (4) find the meeting point of two bodies as the intersection of their graphs.
Skills	Calculating gradient as change in displacement divided by change in time (building on Lesson 3's velocity calculation); reading values from a graph drawn in Lesson 4; analysing and interpreting data (NGSS SEP 4); using mathematics and computational thinking (SEP 5); constructing explanations from graph evidence (SEP 6); communicating and collaborating in pairs.
Attitudes	Appreciating that graphs are a powerful tool for describing and predicting real motion on Kenyan roads; accuracy and patience when reading scales; willingness to justify claims with evidence; respect for road safety.
Key Inquiry Question	1. How do you determine the velocity of a moving body? (Here: from the gradient of its displacement-time graph.)
Purpose in Storyline	In Lesson 4 learners drew displacement-time graphs, and in Lesson 3 they calculated velocity as displacement/time. Lesson 5 turns those graphs into evidence: learners read velocity, direction and rest from them and find where two vehicles meet. This answers the DQB questions 'how do we tell how fast and in which direction?' and prepares for velocity-time graphs in Lessons 6 and 7. A lorry coming towards a car is the same kind of situation as the Naivasha-Nakuru meeting problem, a two-graph meeting, which links to the oncoming lorry L in the phenomenon. Relative speed and safety margins are not calculated here; those come in Lessons 8 and 9.
Safety Notes	This is a paper and graph lesson. Learners remain seated; use sharp pencils safely. If any outdoor walking demonstration of constant speed is used, keep it in a cleared area away from the road. Remind learners never to experiment with overtaking or racing on real roads, and always to cross roads at safe places.

B. LESSON OVERVIEW

Learners start with the overtaking puzzle in mind: how do we describe and predict motion with numbers and graphs? They predict what different graph shapes mean for the returning cyclist Atieno (Naivasha to Lake Naivasha gate and back). In the Observe phase they work in pairs with the graph drawn in Lesson 4, points (0,0), (30,6), (40,6), (60,0) in (min,km), and calculate the gradient of each section: 12 km/h, 0 and -18 km/h. In the Explain phase they use two graphs on the same axes for a matatu leaving Naivasha at 60 km/h and a private car leaving Nakuru at 90 km/h, 60 km apart, and find that they meet after 24 minutes, 24 km from Naivasha. They add new questions to the Driving Question Board and update their model by recording that gradient gives velocity and sign gives direction, and that the oncoming lorry L would have a negative gradient on the same graph as the cars A and B moving towards Nakuru. No new numbers for the overtaking are introduced beyond what learners already know; the speeds of A (20 m/s) and the lorry (-20 m/s) are used only as examples of the sign of the gradient.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Position vs. time graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Position-Time Graphs (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the overtaking clip near Naivasha and the DQB question 'how can we tell how fast and in which direction a vehicle is moving?' The teacher displays the graph for cyclist Atieno drawn in Lesson 4 with points (0,0), (30,6), (40,6), (60,0) in (min,km), with the sections not yet labelled. Individually, learners write a prediction in their notebooks for three questions: (a) Which part shows Atieno riding fastest? (b) What is happening where the line is flat? (c) What does the last part, where the line goes down, mean? They also predict: 'If a matatu leaves Naivasha for Nakuru and a car leaves Nakuru for Naivasha at the same time, where on a graph would we see them meet?' Learners sketch this in 2 minutes and share with a partner. This connects to the oncoming lorry L, which also travels in the opposite direction to the cars.	Display the graph and say: 'You drew this graph last lesson. Do not calculate yet. Look at the three sections and write what you think each one tells us about Atieno's ride.' Give 2 minutes of silent writing. WAIT TIME: wait 10-15 seconds after asking 'What does a flat line mean?' before taking any answer. Cold-call 3 learners with 'Achieng, what did you predict for the middle section? Kiprono, do you agree? Why?' Then ask: 'In the overtaking clip, the lorry L comes towards the cars from Nakuru. If we drew all of them on one graph, how might its line look different from the car's line?' Take 2 more predictions without evaluating. Record predictions on the board under the heading 'Our predictions' to revisit at the end. Say: 'Today we will test these predictions with calculations and graphs.'	Predict-Observe-Explain (Prediction Recording): learners commit to ideas in writing before instruction. This surfaces the common idea that a steeper line means 'higher up' rather than 'faster' and that a falling line means 'slowing down' rather than 'coming back'. Because predictions are written down and revisited, learners see how evidence changes their ideas.	Observe and record: how many learners write that a flat line means 'stopped' or 'not moving' (correct) versus 'moving at steady speed' (misconception)? How many predict that the downward section means 'slowing down' (common misconception) versus 'coming back'? Note whether learners can sketch two lines that cross and say that the crossing shows a meeting. Use a thumbs-up/thumbs-down check on 'a steeper line means faster' and tally the count on the board.
Observe Phase  VIDEO: Interpreting direction of motion from position-time graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Position-Time Graphs Practice Source: CK-12  Search ARES for similar readings	In pairs, learners use the graph of Atieno's ride, points (0,0), (30,6), (40,6), (60,0) in (min,km), and a worksheet. Task 1: For each of the three sections, read the change in displacement and the change in time from the graph and calculate the gradient. Section 1 (0-30 min): 6 km in 30 min = 0.2 km/min = 12 km/h. Section 2 (30-40 min): 0 km in 10 min = 0 km/h (the flat line, resting at 6 km). Section 3 (40-60 min): -6 km in 20 min = -0.3 km/min = -18 km/h. Learners convert km/min to km/h by multiplying by 60. They label each section with a velocity and a direction in words: 'away from home at 12 km/h', 'at rest', 'back	Distribute worksheets and rulers. Say: 'Choose two points on the straight section, find how far up the line goes and how far across, then divide up by across. Show units.' Circulate and ask each pair: 'What does your answer tell us about how fast Atieno is going? In which direction?' WAIT TIME: after asking 'Why is the gradient of the last section negative?' wait 10-15 seconds before calling on anyone. Cold-call 4 pairs: one for each section's gradient and one for the steady car. Ask: 'Which is faster, 12 km/h or -18 km/h? Careful: what does the minus sign tell you and what does the number tell you?' Draw out that the number is	Data Analysis and Pattern Finding with Structured Calculation: learners compute the same quantity (gradient) from three differently-shaped sections and notice the pattern: steeper means faster, upward means positive, downward means negative, flat means zero. Repeating the same calculation on different sections builds a general rule that links the graph to the velocity formula from Lesson 3.	Check worksheets for: gradients 12 km/h, 0 and -18 km/h with units; correct conversion from km/min to km/h; correct labelling of direction in words. Look for learners who compute the gradient as 'time divided by displacement' and prompt them to check the units. Use a quick exit signal: learners hold up fingers 1, 2 or 3 for the section with the greatest speed (answer: 3). Record how many pairs correctly explain that the minus sign shows direction, not 'smaller speed'.

	towards home at 18 km/h'. Task 2: Learners compare the steepness of sections 1 and 3 and say why section 3 is steeper and slopes down. They recall that Atieno's final displacement is 0 and her total distance is 12 km. Task 3: Learners compare with a steady car at 60 km/h, points (0,0), (30,30), (60,60) in (min,km), and check that the gradient is $30 \text{ km} / 30 \text{ min} = 1 \text{ km/min} = 60 \text{ km/h}$. They link the sign of velocity to the oncoming lorry L, whose velocity is -20 m/s because it travels towards Naivasha while the positive direction is towards Nakuru.	the speed and the sign is the direction. Point to the lorry: 'The lorry L has velocity -20 m/s. What would its line on a displacement-time graph do?' Take 2 responses. Address the misconception that a flat line means constant speed by asking: 'If the line is flat, has the displacement changed? So what is the velocity?'		
Explain Phase  VIDEO: Position vs. time graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Position-Time Graphs (Flexbook) Source: CK-12  Search ARES for similar readings	Learners explain how graphs help solve the Naivasha-Nakuru meeting problem, which is similar to the oncoming lorry L approaching the cars. Setting: Naivasha and Nakuru are 60 km apart. A matatu leaves Naivasha for Nakuru at 60 km/h and a private car leaves Nakuru for Naivasha at 90 km/h at the same moment. Taking displacement from Naivasha, learners work out two lines: the matatu (0,0) to (60 min, 60 km), and the car starting at 60 km at 0 min with velocity -90 km/h, reaching (40 min, 0 km) after 40 min. Using a scale such as 1 cm = 5 min and 1 cm = 5 km, they draw both lines on one pair of axes and find the point where the lines cross: at 24 min and 24 km from Naivasha. They check with calculation: matatu $60 \times 0.4 = 24 \text{ km}$; car $90 \times 0.4 = 36 \text{ km}$; $24 + 36 = 60 \text{ km}$. They write a short explanation: 'The two lines cross at 24 min and 24 km because at that moment both vehicles are at the same place at the same time.' Learners then return to the Atieno	Say: 'Two vehicles are at the same place at the same time. What would that look like on a graph?' WAIT TIME: wait 10-15 seconds, then cold-call 3 learners for ideas. Guide the drawing step by step: 'Where does the matatu start? Where does the car start? Which line goes up and which goes down? Why?' Circulate and check each pair's scales. When pairs find the intersection, ask: 'How can we check this answer without the graph?' Cold-call 2 pairs to show the check $60 \times 0.4 = 24$ and $90 \times 0.4 = 36$. Ask: 'In the overtaking clip, the lorry L is coming towards the cars. Would its line go up or down on a graph where up is towards Nakuru? Why?' Cold-call 2 learners. Ask: 'Car A has a steady velocity of 20 m/s. What would its line look like?' Accept 'a straight upward line'. Ask: 'Car B speeds up at first. Predict how the shape of its line would differ.' Do not give the answer: say 'We will test that idea in later lessons when we study velocity and acceleration.' Return to the	Claim-Evidence-Reasoning with Graph Intersection: learners make a claim (the vehicles meet at 24 min and 24 km), support it with evidence from the graph and calculation, and give reasoning (same place, same time). Using two representations, the graph and the arithmetic check, builds understanding that the intersection means the same displacement at the same time. Revisiting the lorry L from the phenomenon transfers the idea to the overtaking puzzle.	Look for: both lines correctly drawn with the car line sloping downwards; intersection read as 24 min and 24 km; calculation check $24 + 36 = 60$; correct statement that the negative gradient means moving towards Naivasha. Listen for the explanation that the intersection shows 'same place, same time'. Note whether learners confuse the intersection with equal speed (misconception) and probe: 'Are the two speeds equal? What is equal at the meeting point?' Use a 3-minute mini-quiz with one problem: 'A graph line goes from (0,0) to (10 min, 5 km) then is flat to 20 min. State the velocity in each section.' Expected: $0.5 \text{ km/min} = 30 \text{ km/h}$, then 0.

	graph and revise their predictions. Finally they explain how a graph could describe the overtake: car A would be a straight line with gradient 20 m/s, car B would curve upward at first because its velocity increases, and lorry L would slope downwards with gradient -20 m/s. They do not calculate relative speed yet.	prediction board and ask learners to circle correct predictions and correct the others using evidence from the graph.		
Driving Question Board (DQB) Creation  VIDEO: Interpreting direction of motion from position-time graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Position-Time Graphs Practice Source: CK-12  Search ARES for similar readings	Learners revisit the Driving Question Board. In groups of four, they write new questions on sticky notes and sort them into categories: 'what we now know', 'what we still need to find out'. Examples of what they now know: 'Gradient of a displacement-time graph gives velocity'; 'a negative gradient means moving in the opposite direction, like the oncoming lorry L'; 'a flat line means at rest'; 'the intersection of two graphs shows where and when two vehicles meet'. Examples of new questions: 'How do we show that Car B speeds up on a graph?', 'What does a graph of velocity against time look like?', 'How can we find the distance from a velocity graph?', 'Why does Car A look slow from Car B when both speedometers show high speeds?', 'How long does the overtaking take?'. Learners place the notes under the two KICD questions: 'How do you determine the velocity of a moving body?' (partly answered today: from the gradient) and 'How do you determine the acceleration of a moving body?' (not yet answered).	Say: 'Look at our Driving Question Board. Which questions can we now answer after today? Move those notes to the answered column and give the evidence.' Cold-call 3 groups to move a note and justify it. WAIT TIME: after asking 'What new question has today's work raised?' wait 10-15 seconds before taking responses. Ask: 'Our graph of the meeting matatu and car shows a straight line for each, because each had a steady velocity. What about Car B, which speeds up? What question does that raise?' Guide learners to the idea that a different kind of graph is needed, without teaching it. Ensure at least 2 new questions point towards velocity-time graphs and 1 towards relative speed. Take a photograph or copy the board for later use.	Question Sorting and Tracking Progress on the DQB: learners categorise questions as answered, partly answered or open, using evidence from the lesson. This shows how each lesson adds one piece to the overall puzzle and motivates the next lesson by making gaps in understanding visible.	Check whether each group moves at least one question with correct evidence, e.g. 'velocity is the gradient' with the Atieno figures 12, 0 and -18 km/h. Look for new questions that go beyond recall, such as asking about acceleration or relative speed. Note groups whose questions show the misconception that the displacement-time graph of an accelerating car is straight; these learners need the follow-up in Lessons 6 and 7.
Model Building Phase  VIDEO:	Learners add to their cumulative model of the overtake. On a page headed 'Our model of the overtaking on the Naivasha	Say: 'Open your model page. Add today's evidence under the new heading and make sure that you use the numbers from Atieno's ride	Cumulative Model Revision: learners add to a growing model, linking each new idea to the anchoring phenomenon. Explicitly	Check model pages for: all four statements recorded with correct figures (12, 0, -18 km/h; 24 min and 24 km); a correct sketch

Interpreting direction of motion from velocity-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Average velocity and average speed from graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	highway', they add a new section called 'Reading displacement-time graphs'. They record: (1) Velocity = gradient = change in displacement / change in time; (2) positive gradient = towards Nakuru (positive direction); negative gradient = towards Naivasha, like the oncoming lorry L with velocity - 20 m/s; (3) flat line = at rest, velocity 0 (like the parked matatu); (4) the meeting point of two bodies is where their lines cross: same time, same place. They draw a small sketch with three lines on one pair of axes: Car A (steady upward line), the lorry L (steady downward line) and a question mark for Car B, noting 'B speeds up, so its line is not straight; we need to find out more.' They then complete a short reflection: 'Today I learned that ... I am still wondering ...'.	and the matatu and car meeting.' Circulate and ask 3 learners: 'How does today's idea help us explain the overtake? What is still missing?' WAIT TIME: wait 10-15 seconds before each answer. Cold-call 3 learners to read their sketch. Ask: 'Why have you put a question mark for Car B?' Accept: 'its velocity changes so its graph would not have a constant gradient'. Close with: 'Next time we will draw a different kind of graph, velocity against time, to show how Car B speeds up.' Collect model pages as a record of progress. Revisit the predictions on the board and mark them right or wrong with learners.	marking what the model cannot yet explain (Car B's speeding up) builds the habit of identifying the limits of a model and motivates the next investigation.	showing lines of opposite gradients; a clear statement of what is still unknown. Use the reflection sentence as an exit ticket and sort into: ready for velocity-time graphs, needs practice on reading gradient with units, needs support on negative gradient. Plan short revision on any group not reaching the expected explanation.
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D. TEACHER REFLECTION

After the lesson, consider: Did learners calculate the gradients 12 km/h, 0 and -18 km/h correctly with units? Which misconceptions were most common (flat line means constant speed; downward line means slowing down; intersection means equal speeds)? Did learners link the negative gradient to the oncoming lorry L? Were scales on the meeting graph chosen sensibly, and who needed support with plotting? Which DQB questions were raised that should be addressed in Lessons 6 to 9? How will you adjust the next lesson to build on learners' readiness to move from displacement-time to velocity-time graphs?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the graph of Atieno's ride has three sections with gradients 12 km/h, 0 and -18 km/h, and that the lines for the matatu leaving Naivasha at 60 km/h and the car leaving Nakuru at 90 km/h cross at 24 min and 24 km from Naivasha.
What did I learn?	We learned that the gradient of a displacement-time graph is velocity, that its sign shows direction (negative means the opposite way, like the oncoming lorry L), that a flat line means rest, and that the intersection of two graphs shows where and when two bodies meet.
How does this explain the phenomenon?	We can explain that the vehicles meet when their graphs cross because they are at the same displacement at the same time. This helps us describe the motion of Car A, Car B and the oncoming lorry L, but we still need velocity-time graphs and acceleration to explain how Car B speeds up when overtaking.

LESSON 6 (80 minutes (2 x 40-minute periods)): Drawing Velocity-Time Graphs: Speeding Up, Steady Speed and Slowing Down

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners plot velocity-time graphs from data tables and recognise the shapes for speeding up (rising line), steady speed (horizontal line) and slowing down (falling line), so that they can describe the motion of Car B, Car A and a braking matatu on the Nairobi-Nakuru highway.
Knowledge	Learners will: (1) state that a velocity-time graph has t (s) on the x-axis and v (m/s) on the y-axis; (2) match a rising line to speeding up (positive acceleration), a horizontal line to steady speed (zero acceleration) and a falling line to slowing down (deceleration, negative acceleration); (3) link the 2 m/s^2 and -2 m/s^2 already calculated in Lesson 3 to the shapes they draw.
Skills	Learners will: choose a scale ($1 \text{ cm} = 1 \text{ s}$ on the x-axis, a suitable scale on the y-axis); plot points from a table; join them with a ruler or a straight edge; label axes with units; describe the shape of each section of a graph in words (SEP 4: Analysing and Interpreting Data; SEP 2: Developing and Using Models).
Attitudes	Learners will show accuracy and neatness when plotting, persevere when a point looks wrong, and cooperate respectfully when checking a partner's graph.
Key Inquiry Question	How do you determine the velocity of a moving body? How do you determine the acceleration of a moving body? (This lesson shows how velocity values from a table become a picture whose shape tells us whether the body is speeding up, moving steadily or slowing down. Gradient and area are interpreted in Lesson 7.)
Purpose in Storyline	Lesson 4 gave learners the skill of plotting displacement-time graphs and Lesson 3 gave the acceleration values (B: 2 m/s^2 ; braking matatu: -2 m/s^2). Lesson 6 turns the velocity data of Car B, Car A and the braking matatu into velocity-time graphs. These graphs are the tool learners will interpret in Lesson 7 (gradient and area) and use in Lesson 8 (relative speed) to explain why A looks slow from B.
Safety Notes	Learners work at desks with rulers and sharp pencils: handle pencils and compasses carefully and keep them away from faces. Use the videos only for seated viewing. When discussing overtaking, remind learners that real overtaking on Kenyan roads is dangerous and that they must never try to copy it, and must always cross roads at safe points.

B. LESSON OVERVIEW

Learners return to the Naivasha overtaking clip. Achieng watches the speedometers from the back seat of Car B, so learners predict what her speedometer readings would look like as a picture. They then plot three velocity-time graphs from tables: Car B ($t = 0$ to 7.5 s , v from 20 to 30 m/s), Car A (constant 20 m/s) and the braking matatu ($t = 0$ to 10 s , v from 20 to 0 m/s). They compare the three shapes, name them as speeding up, steady speed and slowing down, and connect each shape to the acceleration already calculated in Lesson 3. Learners update the DQB and add the graphs to their class model of the overtake.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Interpreting	Learners recall the clip of Wanjiru's white Subaru (Car B) overtaking Mr Kamau's blue Toyota Axio (Car	Display the question: 'If Achieng wrote her speedometer reading every second, what would the	Predict-Observe-Explain (POE): committing to a prediction before seeing the data creates a reason	Observe whether learners (a) put t on the x-axis and v on the y-axis, (b) draw Car A as flat or constant,

change in speed from velocity-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Velocity-Time Graphs (Flexbook) Source: CK-12 Search ARES for similar readings	A) near Naivasha. Achieng, in the back seat of Car B, writes down the speedometer readings: 72 km/h at the start, then higher. Individually, learners sketch on a blank axis (t on x, v on y) what they think a picture of Car A's speed and Car B's speed against time would look like, using two colours. They do not yet have the table. They then pair up and compare sketches.	picture look like for Car A and for Car B?' Say: 'Draw your best idea. There is no wrong prediction today; we will test it.' Circulate and ask 3 pairs: 'What does your line for Car A do, and why?' WAIT TIME 10-15 seconds after each question before accepting an answer. Cold-call 4 learners to describe their sketch for Car B and 2 for Car A. Record two or three different predictions on the board without judging them, e.g. 'B jumps straight up', 'B rises gradually', 'A is flat'.	to examine the evidence, and the gap between prediction and graph exposes ideas such as 'speed changes instantly'.	(c) show Car B higher than or rising from Car A. Note on a checklist the number of learners who draw B as a sudden jump versus a gradual rise to use in the Explain Phase.
Observe Phase VIDEO: Interpreting direction of motion from velocity-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Velocity-Time Graphs (Flexbook) Source: CK-12 Search ARES for similar readings	Learners work in groups of 4 with graph paper, rulers and the three data tables. Table 1, Car B: t = 0, 1, 2, 3, 4, 5, 6, 7, 7.5 s with v = 20, 22, 24, 26, 28, 30, 30, 30, 30 m/s. Table 2, Car A: v = 20 m/s at every time from 0 to 7.5 s. Table 3, braking matatu: t = 0, 2, 4, 6, 8, 10 s with v = 20, 16, 12, 8, 4, 0 m/s. Using the scale 1 cm = 1 s on the x-axis and, for example, 1 cm = 2 m/s on the y-axis, each learner plots Graph 1 (Cars A and B on the same axes, in two colours), then Graph 2 (the braking matatu). They label axes with units (t (s), v (m/s)), join points with a ruler, and write a title for each graph. Learners give each section of each line a word description, e.g. 'rises steadily from 20 to 30 m/s between 0 and 5 s'.	Model the first two points of Car B on the board or projector, thinking aloud: 'The scale is 1 cm for 1 s across. Which scale on the vertical axis lets 0 to 30 m/s fit on the page?' WAIT TIME 10-15 seconds, then cold-call 2 groups for their scale choice. Remind learners of the plotting skills from Lesson 4 by asking: 'What must every axis have?' (label and units). Circulate and check points; when you see an error ask: 'Does this point agree with the table? Check t first, then v.' Ask each group: 'Why is Car A's line the same height every second?' Cold-call 3 learners to explain. Ask a group that finishes early: 'Where does Car B's line stop rising, and what does Achieng's speedometer show after that?'	Data-to-representation modelling (SEP 4 and SEP 2): turning numbers in a table into a picture lets learners see patterns (rising, flat, falling) that are hard to spot in columns, and plotting on the same axes lets them compare A and B directly.	Check each graph for: axes labelled with units; correct scale (1 cm = 1 s); all points plotted within half a small square; straight line drawn with a ruler; Car A horizontal at 20 m/s; Car B rising from 20 m/s at t = 0 to 30 m/s at t = 5 s then horizontal to 7.5 s; matatu falling from 20 m/s at t = 0 to 0 m/s at t = 10 s. Record how many groups need help with scale and with joining points.
Explain Phase VIDEO: Acceleration vs. time graphs <i>Related topic: not an exact match</i>	Groups compare their three lines and complete a sentence frame for each: 'The line is _____, so the vehicle is _____.' They classify: rising line = speeding up; horizontal line = steady speed;	Hold up two learner graphs and ask: 'What is the same and what is different about the lines for Car B and the braking matatu?' WAIT TIME 10-15 seconds, then cold-call 4 learners. Press for evidence:	Claim-Evidence-Reasoning with sentence frames: learners state a claim ('Car B is speeding up from 0 to 5 s'), cite evidence from the graph (rising line from 20 to 30 m/s), and reason using the	Listen for learners who (a) say a horizontal line at 20 m/s means steady speed, not stopped; (b) say the falling line means deceleration and write $a = -2 \text{ m/s}^2$; (c) quote numbers from the graph as

for this lesson. Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Average velocity and average speed from graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	falling line = slowing down. They link each shape to the acceleration from Lesson 3: Car B $a = (30 - 20)/5 = 2 \text{ m/s}^2$ (positive, speeding up), Car A $a = 0$ (steady), matatu $a = (0 - 20)/10 = -2 \text{ m/s}^2$ (deceleration 2 m/s^2). They revisit their Predict sketches and explain what they got right and what changed, for example that Car B reaches 30 m/s gradually over 5 s rather than at once. They also answer: 'On the graph, in which part of the overtake is Car B speeding up and in which part is it steady?' (speeding up for 0-5 s, steady from 5 s to 7.5 s). Finally, they write one sentence about what the graph does NOT yet tell them (how far each car has travelled, covered in Lesson 7).	'Point to the part of the graph that shows that.' Ask: 'Car B's graph is rising and the matatu's graph is falling. How do the signs of the accelerations 2 and -2 match this?' Cold-call 3 learners. Use the everyday words 'speed up' and 'slow down' first and then insist on 'acceleration' and 'deceleration' for formal answers. Correct the common misconception that a horizontal line means the car is stopped: ask 'Car A is on a horizontal line at 20 m/s; is it parked?' and let learners discuss in pairs for 1 minute. Say: 'Before we do gradient and area in the next lesson, just describe the shape.'	acceleration value from Lesson 3. This builds a link between representation and meaning.	evidence. Use an exit-ticket question: 'Sketch the velocity-time shape for a matatu that speeds up, then travels steadily, then slows down.' Look for rising, horizontal, falling in the right order.
Driving Question Board (DQB) Creation VIDEO: Position vs. time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Instantaneous velocity and instantaneous speed from graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)	Learners revisit the class Driving Question Board from Lesson 1 and from the questions about 'how far' and 'how fast'. In groups they sort the questions into: answered by this lesson (e.g. 'Does Car B speed up all the time?'), partly answered (e.g. 'How far do the cars travel?'), and not yet answered (e.g. 'Why does A look slow from B?', 'Was the overtake safe?'). They write one new question on a sticky note arising from the graphs, such as 'What do the gradient and the area under Car B's line tell us?' and place it on the board.	Say: 'Look at the board. Which questions can we now answer using our graphs?' WAIT TIME 10-15 seconds. Cold-call 3 learners to move one sticky note each and explain why. Ask: 'What new question do these graphs make you ask?' Collect at least 4 new questions. Point out that the question 'How far?' needs something more from the graph, which the class will explore in Lesson 7.	Driving Question Board routine: sorting questions by what has been answered keeps the investigation purposeful and shows learners how each piece of evidence moves them towards answering the driving question.	Check that new questions are linked to the graphs and to the phenomenon, for example about distance or relative speed, rather than random. Note how many learners can explain why a question is only partly answered.

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Model Building Phase VIDEO: Interpreting change in speed from velocity-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Horizontal and Vertical Line Graphs (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners add a new panel to the class model of the overtake: a velocity-time graph poster with Car A (horizontal line at 20 m/s) and Car B (rising line from 20 to 30 m/s over 0-5 s, then horizontal to 7.5 s) on the same axes, with the key moments marked: t = 0 B starts to accelerate; t = 5 s B reaches 30 m/s; t = 7.5 s overtaking complete. Beside the graph they write labels such as 'speeding up, $a = 2 \text{ m/s}^2$ ' and 'steady speed, $a = 0$ '. They add a small side panel for the braking matatu labelled 'slowing down, deceleration 2 m/s^2 '. Finally, each learner completes the summary table and writes one sentence on how the model now explains the speed changes of the overtaking car B.	Say: 'Our model must show what the speedometers did. Where on your graph does Wanjiru reach 108 km/h?' Cold-call 2 learners (answer: 30 m/s, at t = 5 s; $30 \times 3.6 = 108 \text{ km/h}$). Ask: 'How does your model explain that both cars look fast on the speedometer?' WAIT TIME 10-15 seconds. Cold-call 3 learners. Ask: 'What can your model not yet tell us?' (distance travelled, relative speed, safety). Ask groups to peer review another group's poster using a 3-point checklist: axes with units, correct shapes, labels.	Cumulative model revision: adding the velocity-time graph to the class model connects the new representation to earlier ideas (velocity, acceleration) and to the phenomenon, and shows which parts of the explanation are still missing.	Check each model for: correct graph of Car A and Car B on the same axes; labelled sections; units; links to the 2 m/s^2 acceleration. Ask 3 learners to explain their model in one minute and look for correct use of 'speeding up', 'steady speed', 'slowing down', 'acceleration' and 'deceleration'.

D. TEACHER REFLECTION

Which learners chose a suitable scale on their own and which needed help? Did any learners treat a horizontal line as 'not moving', and how did the discussion of Car A's steady 20 m/s address this? Were the predicted sketches in the Predict Phase different from the final graphs (for example, did anyone draw B jumping instantly to 30 m/s)? How well did learners connect the shape of each line to the accelerations 2 m/s^2 , 0 and -2 m/s^2 from Lesson 3? What preparation (pre-printed grids, rulers, pencils) would help in the next lesson, where learners interpret gradient and area?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe? Example: Car A's velocity-time graph is a horizontal line at 20 m/s. Car B's graph rises in a straight line from 20 m/s at t = 0 to 30 m/s at t = 5 s and then is horizontal until t = 7.5 s. The braking matatu's graph falls in a straight line from 20 m/s at t = 0 to 0 m/s at t = 10 s.
What did I learn?	What did we learn? Example: A rising line shows speeding up (positive acceleration, e.g. 2 m/s^2), a horizontal line shows steady

	speed (zero acceleration), and a falling line shows slowing down (deceleration, e.g. $a = -2 \text{ m/s}^2$). A velocity-time graph has t (s) on the x-axis and v (m/s) on the y-axis.
How does this explain the phenomenon?	How does this help us explain the overtake? Example: The graphs show that Wanjiru in Car B speeds up gradually from 72 km/h to 108 km/h in 5 s and then holds that speed, while Mr Kamau in Car A stays at 72 km/h. Next we need to find how far each car travels and how their speeds compare.

LESSON 7 (40 minutes): Interpreting Velocity-Time Graphs: Gradient for Acceleration, Area for Distance

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners interpret velocity-time graphs of the overtaking cars and a braking matatu to find acceleration or deceleration from the gradient and distance travelled from the area under the graph, then check the answers against the split-time table.
Knowledge	Gradient of a velocity-time graph = acceleration, $a = (v - u)/t$, in m/s^2 . A negative gradient means deceleration. The area under a velocity-time graph = distance travelled (displacement when velocity does not change sign). Areas are found using a trapezium, rectangle and triangle.
Skills	Calculating gradients from graph points; splitting the area under a graph into a trapezium, rectangle and triangle; calculating areas with units; checking graph results against a data table; communicating reasoning (NGSS SEP: Analyzing and Interpreting Data; Using Mathematics and Computational Thinking).
Attitudes	Accuracy, perseverance, curiosity about road safety, and respect for others' reasoning during discussion.
Key Inquiry Question	1. How do you determine the velocity of a moving body? 2. How do you determine the acceleration of a moving body?
Purpose in Storyline	Lesson 6 gave learners velocity-time graphs drawn from tables for Car B, Car A and the braking matatu. Lesson 5 gave gradient as a rate. Lesson 7 turns those graphs into numbers: B's acceleration of 2 m/s^2 and the distances 200 m (B) and 150 m (A). These distances are needed in Lesson 8 (relative speed and the 50 m gained) and Lesson 9 (the safety margin).
Safety Notes	Learners work seated at desks with rulers, pencils and graph paper. Remind them to keep pencils and rulers on the desk. In the discussion on overtaking, stress that the numbers describe a model, and that real drivers should never overtake without a clear view of the road.

B. LESSON OVERVIEW

Learners return to the Naivasha highway overtaking clip and their Lesson 6 velocity-time graphs for Car B (20 to 30 m/s in 5 s, then steady to 7.5 s) and Car A (steady 20 m/s). They predict what the slope and the shaded region under the line might mean. They find the gradient of B's sloping section (2 m/s^2) and the gradient of the braking matatu's line (-2 m/s^2). They split the area under B's graph into a trapezium (125 m) and a rectangle (75 m), giving 200 m, and the area under A's graph (150 m). They then check these against the split-time table. They add the new evidence to the Driving Question Board (DQB) and revise their model of the overtake.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Interpreting change in speed from velocity-time graph	Learners look at the Lesson 6 velocity-time graph of Car B and Car A on the same axes (1 cm = 1 s on the x-axis). Individually they write for 2 minutes: (a) what does the slope of B's line from 0 to 5 s	Display the graph on the board or on a large chart. Say: "Remember the overtaking clip near Naivasha. Wanjiru in Car B sped up while Mr Kamau in Car A kept steady. Look at the two lines. What might the	Predict-Observe-Explain (prediction stage): making predictions visible lets learners compare their ideas with evidence later, helping them see that graphs carry information, not just shapes.	Look for: learners who connect the steeper line to B speeding up; learners who suggest the area relates to distance (even loosely, e.g. 'the space is how far'); learners who guess B's distance is

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Velocity-Time Graphs (Flexbook) Source: CK-12 Search ARES for similar readings	tell us about the white Subaru driven by Wanjiru? (b) Could the shaded space between the line and the time axis tell us how far B travelled? Pairs share predictions and write a number guess for B's distance in 7.5 s and A's distance in 7.5 s, using what they know from the clip.	slope be telling us? What might the space under each line be telling us? Write your idea before you speak." Give WAIT TIME of 10-15 seconds before taking any answer. Cold-call 4 learners (2 for slope, 2 for area) and record predictions on a chart headed 'Our predictions'. Do not confirm or correct. Say: "Today we test these ideas with evidence."	It links directly to the phenomenon of how long and how far the overtake takes.	more than A's. Note on a checklist which learners predicted distance and which were unsure, to guide questioning later.
Observe Phase VIDEO: Interpreting change in speed from velocity-time graph Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Velocity-Time Graphs (Flexbook) Source: CK-12 Search ARES for similar readings	Part 1, gradient: in pairs, learners use their own Lesson 6 graph of Car B. They mark two points on the sloping section, (0, 20) and (5, 30), draw a right-angled triangle and calculate gradient = change in v / change in t = $(30 - 20)/5 = 2 \text{ m/s}^2$. They calculate the gradient of the flat sections: Car A from 0 to 7.5 s and Car B from 5 to 7.5 s, getting 0. They then take the braking matatu graph (points (0, 20), (2, 16), (4, 12), (6, 8), (8, 4), (10, 0)) and calculate gradient = $(0 - 20)/10 = -2 \text{ m/s}^2$, writing 'deceleration 2 m/s^2'. Part 2, area: learners shade the region under B's line and split it into a trapezium from 0 to 5 s with area $\frac{1}{2} \times (20 + 30) \times 5 = 125 \text{ m}$ and a rectangle from 5 to 7.5 s with area $30 \times 2.5 = 75 \text{ m}$, total 200 m. They calculate A's rectangle: $20 \times 7.5 = 150 \text{ m}$. They calculate the matatu's triangle: $\frac{1}{2} \times 20 \times 10 = 100 \text{ m}$ stopping distance. Part 3, check: learners compare with the split-time table (B distance 200 m and A distance 150 m at 7.5 s; B at 5 s = 125 m, matching the trapezium). They also find B minus A = 50 m.	Circulate with a checklist. Say: "Show me the two points you chose. Which one is (t, v)? What are the units of your answer?" Prompt: "Why is the gradient of Car A's line zero? What does that say about its acceleration?" For the matatu: "Your answer is negative. What does the negative sign tell us about the matatu?" Insist on 'deceleration 2 m/s^2 means $a = -2 \text{ m/s}^2$'. For area: "Which shape fits 0 to 5 s? Which fits 5 to 7.5 s? Name them before you calculate." Stop the class after the trapezium and ask for WAIT TIME of 10 seconds: "How can we check 125 m without using the graph?" Direct learners to the split-time table. Cold-call 3 pairs to report their totals (B 200 m, A 150 m, matatu 100 m). Support slower learners with a pre-drawn shaded graph; extend faster learners by asking them to find the area under A's line from 0 to 5 s (100 m) and compare with the table.	Data Analysis with Triangulation: learners find the same quantity two ways (graph area and the split-time table), which builds confidence that the area under a velocity-time graph is distance, and that the gradient is acceleration. This answers the KICD inquiry question on how to determine acceleration.	Observable indicators: correct gradient 2 m/s^2 with units; correct sign -2 m/s^2 for the matatu; correct split of B's area into a trapezium and a rectangle; correct totals $125 + 75 = 200 \text{ m}$, 150 m, 100 m; learners naming the match with the table without being told. Common errors to note: dividing time by velocity change, forgetting the $\frac{1}{2}$ in the trapezium or triangle formula, using 7.5 instead of 2.5 for the rectangle width, and reporting a negative distance.
Explain Phase VIDEO: Acceleration vs.	In groups of 4 learners prepare a short explanation titled 'What the graph tells us about the overtake':	Say: "Wanjiru's friend Achieng watched the speedometer from the back seat of Car B. Use your	Claim-Evidence-Reasoning (CER): each group states a claim (e.g. B travels 200 m while A travels 150	Indicators: groups explain that a sloping line means changing speed and a flat line means steady

time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Average velocity and average speed from graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	(1) Why does B's line slope up for 5 s and then go flat? (2) Why is B's distance (200 m) bigger than A's (150 m), and what does the 50 m difference represent in the real road (20 m starting gap + 5 m + 5 m car lengths + 20 m finishing gap)? (3) How do we know from the matatu graph that it slows down at a steady rate? Each group writes one sentence for gradient and one for area using the frame 'The gradient of the line is ... which means ...' and 'The area under the line is ... which means ...'. Learners also answer: what would the line for an oncoming lorry L look like at -20 m/s, and why would its area need care (direction)?	graph to explain what she saw from 0 to 5 s and from 5 to 7.5 s." Give WAIT TIME of 10-15 seconds, then cold-call 3 groups. Probe with: "How do you know?", "What evidence in the table agrees?" and "What does 2 m/s^2 mean in words?" (speed increases by 2 m/s every second: 20, 22, 24, 26, 28, 30). Address the misconception that a bigger speed means larger acceleration: ask "Car A is at 20 m/s and B is at 30 m/s later. Which has the greater acceleration at 6 s?" (both zero). Link the 50 m to the milestone that B is level with A's front at 5 s (gained 25 m) and complete at 7.5 s (gained 50 m). Say: "Next lesson we will explain why A looks slow from B using relative speed."	m), supports it with evidence (areas and the split-time table) and gives reasoning (area under a v-t graph equals distance). This builds a scientific habit of justifying numbers and ties the mathematics back to the overtaking phenomenon.	speed; groups connect 2 m/s^2 to 'gains 2 m/s each second'; groups state area = distance with units; groups justify the 50 m difference with the gap and car lengths. Use a 3-level checklist: states the number only; states the number with meaning; states the number, meaning and evidence from the table.
Driving Question Board (DQB) Creation VIDEO: Interpreting direction of motion from velocity-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Instantaneous velocity and instantaneous speed from graphs <i>Related topic: not an exact match for this lesson.</i>	Learners revisit the DQB from Lesson 1. Using sticky notes they add (a) answers to earlier questions about 'how far' and 'how fast', e.g. 'B travels 200 m and A travels 150 m in 7.5 s (areas under graphs)', and 'B accelerates at 2 m/s^2 (gradient)'; (b) new questions, e.g. 'Why does A look slow from B if both speedometers read high?', 'Is there enough road before the lorry arrives?'. Each learner places at least one note and explains it to a neighbour.	Say: "Look at our board. Which questions can we now answer with graph evidence? Move the note and write the evidence on it." WAIT TIME 10 seconds. Cold-call 3 learners to place and justify notes. Point out the remaining questions on relative speed and safety, and mark them as 'Lesson 8' and 'Lesson 9'. Remind the class of the two KICD questions: "How do you determine the velocity of a moving body? How do you determine the acceleration of a moving body?" and ask which part of acceleration they now know.	Driving Question Board tracking: showing questions answered and questions still open makes the learning journey visible and gives purpose to the next lessons, keeping learners focused on the driving question about describing and predicting the overtake.	Indicators: sticky notes include evidence (a number or graph feature) and not just a statement; new questions are linked to relative speed or safety; learners can say which earlier question the new evidence answers. Record the number of learners who linked evidence to questions.

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Model Building Phase VIDEO: Position vs. time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Decompose area with triangles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners update their class overtaking model (annotated diagram plus graphs). Individually they add to their model: (1) B's v-t graph labelled with 'gradient = 2 m/s ² ' on 0-5 s and 'gradient = 0' on 5-7.5 s; (2) 'area = 125 m + 75 m = 200 m'; (3) A's graph labelled '20 m/s, area = 150 m'; (4) a note 'B gains 200 - 150 = 50 m'. They add a small inset for the braking matatu: 'a = -2 m/s ² , stopping distance 100 m'. Then they write a two-sentence statement on how gradient and area help describe and predict the overtake, and mark one idea they are still unsure of.	Say: "Take out your model from Lesson 6. Add what we have learned today. Use different colours for gradient and area." Give WAIT TIME of 10 seconds before asking learners to start writing. Circulate and check that units are present (m/s ² , m). Cold-call 2 learners to read their two-sentence statements. Close by saying: "We know B travels 200 m and A 150 m. In Lesson 8 we ask: how fast does A seem to move from B, and what happens with the oncoming lorry L?" Collect models for review.	Cumulative Model Revision: adding gradient and area to the model converts a picture of the overtake into a quantitative tool. Learners see how each new piece of evidence improves their ability to describe and predict motion.	Indicators: model contains correct labels for acceleration (2 m/s ²), distances (200 m, 150 m), gain (50 m) and the matatu (-2 m/s ² , 100 m); units are included; learners state at least one remaining uncertainty. Use the exit statement to decide which learners need support with trapezium areas or the sign of acceleration at the start of Lesson 8.

D. TEACHER REFLECTION

Which learners could find a gradient from two points and explain what it means without prompting? Who confused the gradient (acceleration) with the height of the line (velocity)? How many learners split B's area into a trapezium and a rectangle by themselves? Were the table checks convincing? Did learners show that deceleration 2 m/s² means a = -2 m/s²? What will I change in Lesson 8 to build from the 50 m gain to relative speed?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that Car B's velocity-time line slopes upward from 20 m/s to 30 m/s in 5 s, then stays flat to 7.5 s; Car A's line is flat at 20 m/s; the braking matatu's line slopes downward from 20 m/s to 0 in 10 s. The areas under the graphs matched the split-time table.
What did I learn?	We learned that the gradient of a velocity-time graph gives acceleration (B: 2 m/s ² ; matatu: -2 m/s ² , deceleration 2 m/s ²) and the area under the graph gives distance travelled (B: 125 + 75 = 200 m; A: 150 m; matatu: 100 m).

How does this explain the phenomenon?

We can explain that Car B travels 50 m more than Car A during the overtake because it sped up at 2 m/s^2 , and that velocity-time graphs let us describe and predict the motion of the cars on the Naivasha highway using gradient and area.

LESSON 8 (40 minutes (one lesson)): Relative Speed: Why Does the Overtaken Car Look Slow?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners determine the relative speed of two bodies moving in the same direction (subtract speeds) and in opposite directions (add speeds), always stating the viewpoint, and use it to explain why overtaken car A looks slow from overtaking car B even though both speedometers read high.
Knowledge	Relative speed of two bodies moving in the same direction is the difference of their speeds; in opposite directions it is the sum of their speeds. The viewpoint must be stated (e.g. 'B relative to A'). B relative to A grows from 0 m/s at $t = 0$ to 10 m/s (36 km/h) at $t = 5$ s and then stays at 10 m/s. B and oncoming lorry L close at $30 + 20 = 50$ m/s (180 km/h). The extra distance B gains on A (1, 4, 9, 16, 25, 35, 45, 50 m) is the area under B's relative velocity graph: $\frac{1}{2} \times 5 \times 10 = 25$ m plus $10 \times 2.5 = 25$ m, total 50 m.
Skills	Calculating relative speed; converting between km/h and m/s (divide or multiply by 3.6); reading the B-minus-A column of the split-time table; drawing and interpreting a relative velocity-time graph; finding area under a graph (triangle and rectangle); communicating reasoning from a stated viewpoint. NGSS SEPs: Analyzing and Interpreting Data; Using Mathematics and Computational Thinking; Constructing Explanations.
Attitudes	Curiosity about how everyday road observations can be explained with numbers; precision in stating viewpoints and units; responsibility and road safety awareness when thinking about overtaking on Kenyan roads; respect for others' reasoning during discussion.
Key Inquiry Question	1. How do you determine the velocity of a moving body? 2. How do you determine the acceleration of a moving body? (Lesson 8 extends question 1: velocity from one body's viewpoint is found by combining the velocities of two bodies; the acceleration of B from Lesson 3 explains why relative speed grows from 0 to 10 m/s.)
Purpose in Storyline	Lessons 3 and 7 gave the velocities of A (20 m/s), B (20 to 30 m/s, $a = 2 \text{ m/s}^2$) and L (-20 m/s) and the distances from areas under graphs. Lesson 8 answers the puzzle from Lesson 1: how can two fast vehicles look slow relative to each other? It supplies the relative speeds (10 m/s and 50 m/s) that Lesson 9 will use to judge whether the overtake is safe.
Safety Notes	This is a classroom paper-and-graph lesson. If learners act out relative motion in the corridor or field, walk slowly in marked lanes, keep clear of doors and roads, and never run towards each other. Remind learners never to judge overtaking distances on a real road from a classroom rule; Kenyan road safety rules and a driver's full attention always apply.

B. LESSON OVERVIEW

Learners return to the Naivasha highway clip: both speedometers show high speeds (72 km/h and up to 108 km/h), yet car A looks slow from car B. They first predict relative speeds, then gather evidence from the split-time table (B minus A column: 0, 1, 4, 9, 16, 25, 35, 45, 50 m) and by calculating B relative to A and B and L closing speeds. They explain the results with a relative velocity-time graph whose area equals the 50 m gained, update the Driving Question Board with new questions about safety, and revise their class model, adding relative speed as a link between the speedometers and what the drivers see.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Finding Speed when Objects Travel in Opposite Directions Source: CK-12  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: linear motion relative speed	Learners re-watch the Naivasha clip (or hear it described) and look at the speedometer facts at $t = 0$: A and B both show 72 km/h (20 m/s). In pairs they predict and write: (a) How fast does A seem to move from Wanjiru's seat in car B at $t = 0$? (b) How fast does A seem to move when B has reached 108 km/h? (c) How fast does oncoming lorry L (72 km/h towards Naivasha) seem to approach car B when B is at 108 km/h? They record the prediction with a short reason in their notebooks and draw a quick sketch with arrows for direction.	Say: 'Imagine you are Achieng in the back seat of car B watching car A through the windscreen. Does A look fast or slow? Write your prediction before we calculate anything.' Give 2 minutes of individual writing, then WAIT TIME of 10-15 seconds after asking: 'What do you think happens to how fast A seems to move as B speeds up?' Cold-call 4 pairs and record predictions on the board without judging them as right or wrong. Say: 'Hold on to your prediction; we will test it with the numbers from the split-time table.'	Predict-and-record (Think-Pair-Share): committing to a prediction in writing makes learners aware of their prior ideas, such as 'a fast car must look fast', so that the later evidence creates real conceptual conflict and is remembered.	Look for: predictions written down by every learner; sketches with direction arrows; reasons that mention viewpoint or speed difference. Note how many of the 4 cold-called pairs predict that A looks fast from B at $t = 0$ (a common misconception) and tally predictions on the board for comparison at the end of the lesson.
Observe Phase  VIDEO: Finding Speed when Objects Travel in Opposite Directions Source: CK-12  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: linear motion relative speed	Learners work in groups of 4 with the split-time table. Step 1: Study the B minus A column (0, 1, 4, 9, 16, 25, 35, 45, 50 m at $t = 0, 1, 2, 3, 4, 5, 6, 7, 7.5$ s) and calculate how much extra distance B gains each second: 1, 3, 5, 7, 9 in the first five seconds, then 10 m per second after $t = 5$ s (25 to 35 to 45). Step 2: Calculate B relative to A as the difference of velocities at each time: $t = 0, 0$ m/s; $t = 1, 22 - 20 = 2$; $t = 2, 4$; $t = 3, 6$; $t = 4, 8$; $t = 5, 10$ m/s; and 10 m/s afterwards. Convert 10 m/s to 36 km/h. Step 3: Calculate the closing speed of B and oncoming lorry L when B is at 30 m/s: $30 + 20 = 50$ m/s = 180 km/h. Step 4: Role play in the corridor with two learners walking in the same direction at different slow speeds, then towards each other, noticing how fast the gap changes.	Distribute the table and circulate. Say: 'Look at the B minus A column. What does the gap-gaining pattern tell you about how fast B is moving relative to A?' Prompt with: 'At $t = 0$ both speedometers show 72 km/h. What is B relative to A at that moment?' WAIT TIME of 10-15 seconds before taking answers. Ask: 'Why do we subtract the speeds for A and B but add them for B and L?' and cold-call 3 groups. Remind groups to write the viewpoint each time, e.g. 'B relative to A = $30 - 20 = 10$ m/s'. Use the corridor demonstration with two learners: walking the same way at 1 m/s and 2 m/s, then walking towards each other, and ask the class to describe the speed at which the gap changes.	Data pattern analysis with a structured table: learners compute a new column (relative speed) from given data and look for patterns, which builds the rule 'same direction subtract, opposite direction add' from evidence rather than being told it. The physical walking demonstration gives a concrete, embodied model of the rule.	Check that groups: calculate B relative to A as 0, 2, 4, 6, 8, 10, 10 m/s correctly; write the viewpoint in every statement; use 72 km/h = 20 m/s and 108 km/h = 30 m/s correctly; compute 50 m/s and 180 km/h for B and L. Common errors to watch: subtracting for opposite directions, forgetting units, or giving 100 m/s because both are written as positive speeds. Tick off each group on a checklist as they complete each step.
Explain Phase  VIDEO:	Learners use their calculations to build a written explanation in the	Say: 'Use the sentence frame: From the viewpoint of ____, the	Claim-Evidence-Reasoning with a graphical model: learners link a	Look for explanations that (1) state the viewpoint, (2) subtract for

Interpreting direction of motion from velocity-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: linear motion relative speed	format Claim-Evidence-Reasoning. Claim: car A looks slow from car B because their relative speed is small. Evidence: at $t = 0$ B relative to A = 0 m/s; after $t = 5$ s it is only 10 m/s (36 km/h); B gains 50 m in 7.5 s. Reasoning: both cars move in the same direction so we subtract speeds. They then draw a velocity-time graph of B relative to A (t on the x-axis, 1 cm = 1 s; relative velocity on the y-axis) rising as a straight line from 0 to 10 m/s at 5 s, then flat to 7.5 s. They find the area: triangle $\frac{1}{2} \times 5 \times 10 = 25$ m plus rectangle $10 \times 2.5 = 25$ m, total 50 m, and match it to the last row of the B minus A column (50 m = 20 m gap + 5 m + 5 m + 20 m). They also explain why oncoming lorry L is dangerous: closing speed 50 m/s (180 km/h).	other vehicle seems to move at ____ because ____.' Ask: 'The speedometer says 108 km/h in car B, yet A looks slow. How can both be true?' WAIT TIME of 10-15 seconds, then cold-call 3 learners to share their reasoning. Ask: 'Which slope on your relative graph tells us B's relative speed is growing? What does the gradient 2 m/s² from Lesson 3 have to do with it?' Guide learners to see that the gradient of the relative graph equals B's acceleration because A has no acceleration. Ask: 'Why does the area under the relative graph give 50 m, and where have you seen 50 m before?' Ask learners to compare to their predictions from the Predict Phase.	verbal explanation to numbers and to a graph; using the area under the relative velocity graph connects Lesson 7's area idea with the B minus A column, which strengthens the link between multiple representations (table, graph, words).	same direction and add for opposite, (3) use the 10 m/s (36 km/h) and 50 m/s (180 km/h) values correctly, and (4) show the graph with correct axes, labels and scale (1 cm = 1 s). Check that the area calculation gives $25 + 25 = 50$ m. Ask 5 learners to explain one step aloud; note who can explain why relative speed starts at 0 when both speedometers show 72 km/h.
Driving Question Board (DQB) Creation VIDEO: Position vs. time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: linear motion	Learners revisit the Driving Question Board. They sort existing question cards into 'answered', 'partly answered' and 'still open'. They place answered cards such as 'Why does A look slow from B?' with a note: 'B relative to A is only 10 m/s (36 km/h) in the same direction; it was 0 at $t = 0$.' They write new questions on sticky notes, e.g. 'Is 150 m of space left when B pulls in enough?', 'How much road does B need in total with lorry L approaching?', 'What if the lorry had been closer?' and 'How do we know the total time of 7.5 s is safe?'	Say: 'Walk to the board in your groups and decide which of our questions we can now answer using relative speed.' Ask: 'Which questions are still open, and which of them need us to combine distance, time and relative speed?' WAIT TIME of 10-15 seconds before taking suggestions. Cold-call 3 groups to read one answered and one new question. Point out the new safety questions and say: 'These will lead us into the next lesson, where we decide whether the overtake is safe.' Add a card: 'How can we use relative speed to describe and predict the overtake?'	Driving Question Board tracking: sorting and generating questions makes learning progress visible, links this lesson's evidence to the main driving question, and gives learners ownership of what to investigate next.	Check that at least 3 new learner questions refer to relative speed, total road needed or lorry L, and that answered cards include evidence (a number and the viewpoint). Observe whether learners can explain why a card moves from 'open' to 'answered'. Note groups who write only vague questions such as 'How fast?' and prompt them to add a viewpoint or unit.

relative speed				
Model Building Phase  VIDEO: Interpreting direction of motion from position-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: linear motion relative speed	Learners revise the class model of the overtaking (a labelled diagram with a timeline and graphs). They add: (1) arrows for A (20 m/s), B (20 to 30 m/s) and L (-20 m/s) with positive direction towards Nakuru; (2) a 'relative speed' box: B relative to A = 0 to 10 m/s (36 km/h) by subtraction; B and L closing speed 50 m/s (180 km/h) by addition; (3) the relative velocity-time graph with the area 50 m; (4) a short caption: 'A looks slow from B because the relative speed is only 10 m/s, even though both speedometers read high.' They then compare with their initial predictions, correct any that were wrong and write one sentence on what changed in their thinking.	Say: 'Update your model so that someone who was not in class could understand why A looks slow from B.' Circulate and ask: 'Where on your model can I see the viewpoint? Where is the 50 m? Which part shows the lorry closing at 180 km/h?' WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to present one addition to the model. Ask: 'What does the model not yet tell us?' to lead towards the safety margin in Lesson 9. Return to the tally of predictions from the Predict Phase and ask learners to say how their ideas changed.	Cumulative model revision: adding relative speed to the existing model lets learners integrate new knowledge with earlier ideas (velocity, acceleration, graphs, areas) and see how the parts explain the phenomenon as one system.	Look for models that include: viewpoint statements, correct values 10 m/s and 50 m/s with km/h conversions (36 km/h and 180 km/h), the 0 to 10 m/s growth, the 50 m gained, and a caption explaining why A looks slow. Use a 3-point checklist (viewpoint, rule for same/opposite direction, correct numbers) and record how many groups meet all three. Exit ticket: 'Write one sentence that explains why A looks slow from B.'

D. TEACHER REFLECTION

Which learners still believed that 'a fast speedometer reading means the other car must look fast', and what evidence changed their mind? Did learners state the viewpoint every time (e.g. 'B relative to A')? Did anyone add speeds for the same direction or subtract them for opposite directions, and how did the corridor demonstration or the graph help correct this? Were the km/h and m/s conversions secure? Which questions on the DQB show learners are ready to combine time, distance and relative speed in Lesson 9? Note any adjustments to pacing, grouping or materials for next time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that at $t = 0$ both car A and car B show 72 km/h (20 m/s), yet B gains 1, 4, 9, 16, 25, 35, 45 and 50 m on A at $t = 1, 2, 3, 4, 5, 6, 7$ and 7.5 s, and oncoming lorry L moves towards them at 20 m/s.
What did I learn?	We learned that relative speed is found by subtracting speeds for bodies moving in the same direction (B relative to A = $30 - 20 = 10$ m/s = 36 km/h, growing from 0 at $t = 0$ to 10 m/s at $t = 5$ s) and by adding speeds for bodies moving in opposite directions (B and L: $30 + 20 = 50$ m/s = 180 km/h), always stating the viewpoint. The area under B's relative velocity graph, $25 + 25 = 50$ m, gives the extra distance B gains.

How does this explain the phenomenon?

Car A looks slow from car B because they move in the same direction, so their relative speed is only 10 m/s (36 km/h) and was 0 at the start, even though both speedometers read high. The oncoming lorry looks very fast from B because it closes at 50 m/s (180 km/h), which is why the overtake needs careful checking in the next lesson.

LESSON 9 (80 minutes (2 x 40-minute periods)): Applying Linear Motion: Is Wanjiru's Overtake Safe?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners combine time, distance and relative speed to decide whether the overtake on the Nairobi-Nakuru highway is safe, using the total road needed (350 m) and the safety margin in metres and seconds.
Knowledge	Learners will be able to: (1) state that B needs 7.5 s to complete the overtake, during which B travels 200 m, A travels 150 m and oncoming lorry L travels 150 m; (2) explain that B and L together use $200 + 150 = 350$ m of the road; (3) find the safety margin as $500 - 350 = 150$ m and convert it to time using closing speed 50 m/s ($150 / 50 = 3.0$ s); (4) apply the same method to the what-if case of a lorry 400 m away (margin 50 m, 1.0 s, unsafe).
Skills	Calculating with $a = (v - u)/t$ and areas under a velocity-time graph (Lesson 7); using relative speed from a stated viewpoint (Lesson 8); organising multi-step calculations in a table; making and justifying a safety decision with evidence (NGSS SEP: Using Mathematics and Computational Thinking; Engaging in Argument from Evidence).
Attitudes	Learners show responsibility and care for road safety, respect for evidence-based decisions, and curiosity and perseverance when combining several ideas to solve a real problem.
Key Inquiry Question	1. How do you determine the velocity of a moving body? 2. How do you determine the acceleration of a moving body? (Learners use both, $v = \text{displacement}/\text{time}$ and $a = (v - u)/t$, as tools in the safety decision.)
Purpose in Storyline	Lessons 3, 7 and 8 gave us the tools: velocity and acceleration, areas under velocity-time graphs, and relative speed. Lesson 9 applies them to answer the real question behind the phenomenon: can Wanjiru overtake before the oncoming lorry arrives? It prepares learners for Lesson 10, where they present the full model and answer the driving question.
Safety Notes	All work is classroom-based, using video, paper and calculators. Do NOT ask learners to measure speeds on a real road. Use the lesson to discuss safe overtaking: never overtake when an oncoming vehicle is close, on bends or on hills. Move desks aside for the role-play and keep walkways clear.

B. LESSON OVERVIEW

Learners revisit the overtaking clip near Naivasha and predict whether Wanjiru's Subaru (Car B) can overtake Mr Kamau's Axio (Car A) safely when Mr Otieno's lorry (L) is 500 m ahead. They rebuild the evidence from earlier lessons (B: 20 to 30 m/s in 5 s, $a = 2 \text{ m/s}^2$; B travels 200 m, A 150 m, L 150 m in 7.5 s; relative speed 10 m/s for B relative to A and 50 m/s closing speed for B and L). They then calculate the total road needed (350 m), the safety margin (150 m, 3.0 s) and test a what-if where the lorry is only 400 m away (margin 50 m, 1.0 s, unsafe). The lesson closes by adding the safety decision to the DQB and to the class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Acceleration vs.	Learners re-watch the Naivasha overtaking clip (or view the teacher's description card) and	Play the clip twice, pausing at $t = 0$ when both cars show 72 km/h. Say: 'Wanjiru is about to speed up.'	Predict-Share-Revise: committing to a written prediction and a list of needed information exposes	Look for: (a) learners who write a Yes/No/Not sure decision with a reason; (b) whether the list of

time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: linear motion overtaking relative	individually write a prediction in their notebooks: 'Is Wanjiru's overtake safe when the oncoming lorry is 500 m from B at $t = 0$? What do we need to know to decide?' They sketch a simple road diagram showing B, A and L, then pair up and compare. They list which quantities they think matter (time, distances, speeds).	Mr Otieno's lorry is 500 m ahead, coming towards us. Would YOU pull out? Write Yes, No or Not sure, and one reason.' WAIT TIME: 15 seconds of silent writing. Then cold-call 4 learners from different parts of the room: 'Tell us your decision and what information you would want.' Record the wanted information on the board as a list (time to overtake, distance of B, distance of lorry, speeds). Do not confirm any answer yet. Say: 'By the end of today we will decide using numbers, not feelings.'	preconceptions (e.g. 'fast cars are always safe', 'only B's speed matters'). Learners will later revise their prediction against calculated evidence, making the change in their thinking visible.	needed information includes the lorry's speed and the time or distance of the overtake; (c) misconceptions such as 'only B's speed matters' or 'the lorry is far, so it is safe'. Note names on a checklist for targeted questioning later.
Observe Phase VIDEO: Interpreting direction of motion from velocity-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: linear motion overtaking relative	Working in groups of four, learners complete an evidence table using the data from earlier lessons and the split-time table. Roles: Reader, Calculator, Recorder, Checker. Step 1: from the v-t graph for B, find the acceleration $(30 - 20)/5 = 2 \text{ m/s}^2$ and areas: trapezium $(20 + 30)/2 \times 5 = 125 \text{ m}$ plus rectangle $30 \times 2.5 = 75 \text{ m}$, giving B = 200 m in 7.5 s. Step 2: A travels $20 \times 7.5 = 150 \text{ m}$; lorry L travels $20 \times 7.5 = 150 \text{ m}$ (velocity -20 m/s, distance is a magnitude). Step 3: check the relative distance B minus A = $200 - 150 = 50 \text{ m} = 20 \text{ m gap} + 5 \text{ m (A length)} + 5 \text{ m (B length)} + 20 \text{ m finishing gap}$. Step 4: calculate total road used by B and L together: $200 + 150 = 350 \text{ m}$. Step 5: find the road left: $500 - 350 = 150 \text{ m}$. Step 6: using closing speed $30 + 20 = 50 \text{ m/s}$, convert to time: $150 / 50 = 3.0 \text{ s}$. They label a road diagram at $t = 0$ and $t = 7.5 \text{ s}$ with these distances.	Distribute the split-time table and a blank evidence table. Circulate and ask: 'Which lesson's tool are you using here, velocity, areas or relative speed?' and 'Why do we ADD the distances of B and L but SUBTRACT the distances of B and A?' WAIT TIME: 10 seconds after each question before taking answers. For groups stuck on Step 4, prompt: 'Both vehicles are using the same strip of road, coming from opposite ends. Draw them at $t = 0$ and $t = 7.5 \text{ s}$. How much of the 500 m have they eaten up together?' After 15 minutes, cold-call 3 groups to report one row each of the table. Check the 7.5 s row of the split-time table (B 200 m, A 150 m, B minus A = 50 m) against learners' answers. Remind them the safe-overtake calculation uses only the numbers established in the clip; no other values are to be invented.	Structured Data Table with Role Rotation: breaking the problem into labelled steps lets learners connect three prior tools (acceleration, area under the v-t graph, relative speed) into one chain of reasoning, and the diagram links abstract numbers to the physical road near Naivasha. Cross-checking with the split-time table builds confidence in the model.	Observable indicators: learners correctly get $125 + 75 = 200 \text{ m}$; state A = 150 m and L = 150 m; add rather than subtract to get 350 m; compute 150 m and 3.0 s with units; and can explain 50 m as $20 + 5 + 5 + 20$. Common errors to note: using 72 km/h without conversion, using -20 m/s in a distance sum (giving 50 m), dividing 150 by 30 or 20 instead of 50, forgetting the car lengths.
Explain Phase	Each group writes a 'Safety Verdict' on a half sheet of paper	Model the CER structure once with a sentence frame on the board: 'I	Claim-Evidence-Reasoning with a what-if (variable change):	Check verdicts for: a clear claim; use of 150 m and 3.0 s as

 VIDEO: Position vs. time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: linear motion overtaking relative	using the claim-evidence-reasoning format: Claim (Is the overtake safe with the lorry 500 m away?), Evidence (350 m used, 150 m left, 3.0 s), Reasoning (why the margin is positive and what 3.0 s means for Mr Otieno and Wanjiru). Then the class tests the what-if: the lorry is only 400 m away at $t = 0$. Groups calculate: road used is still 350 m, margin $400 - 350 = 50$ m, time $50 / 50 = 1.0$ s, and decide this is unsafe, explaining that a 1.0 s gap is far too small for drivers to react or brake. They also explain, in their own words, why both cars can show high speedometer readings yet A looks slow from B (relative speed 10 m/s, 36 km/h), while the lorry seems very fast from B (50 m/s, 180 km/h). They revisit their Predict Phase decision and revise it.	claim ____ because the evidence shows ____, which means ____.' Then say: 'Now change one number. The lorry is 400 m away. Predict first: safe or unsafe? Show with fingers.' WAIT TIME: 10 seconds, then ask groups to calculate. Cold-call 5 learners to give one step each of the what-if solution: road used, margin in metres, closing speed, margin in seconds, verdict. Ask: 'Why does the 350 m not change when the lorry is closer?' and 'What would you say to a driver who says 1.0 s is enough?' WAIT TIME 15 seconds before taking responses. Stress the conventions: say 'overtaking car B', 'overtaken car A' and 'oncoming lorry L', and always state the viewpoint when using relative speed. Link to the key inquiry questions: ask which calculation used velocity and which used acceleration.	changing only one variable (lorry distance) isolates cause and effect, helping learners see that safety depends on the margin, not on how fast the cars look. Writing reasoning in sentences forces them to connect numbers to the physical meaning of the overtake.	evidence; the what-if numbers 50 m and 1.0 s; the explanation that 1.0 s is unsafe. Success indicators: learners use 'relative speed of B relative to A is 10 m/s' and 'closing speed of B and L is 50 m/s' correctly with viewpoint. Ask 3 learners to restate the verdict without notes. Misconception check: 'Because B is faster than L, there is no danger.'
Driving Question Board (DQB) Creation  VIDEO: Acceleration vs. time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES	Learners return to the class Driving Question Board started in Lesson 1. In their groups, they pick questions already on the board about 'how long does it take', 'how far do the cars travel' and 'is it safe' and mark which ones they can now answer with evidence (7.5 s; B 200 m, A 150 m, L 150 m; 350 m used; margin 150 m or 3.0 s). They write answers on sticky notes with the data source (for example, 'area under v-t graph, Lesson 7'). They add new questions, such as 'What if B accelerated faster?', 'What if the road were wet?', 'What if the lorry was heavier or faster?', and 'What is the minimum safe distance for the lorry?' and place them on the board.	Say: 'Let us look at our board. Which questions can we now answer, and which have we only partly answered?' Ask 4 learners (cold-call) to bring a note to the board and read their answer with the evidence. WAIT TIME: 10 seconds before asking, 'What new question did today raise for you?' Group questions by type: calculation, safety, and beyond-the-data (road conditions, driver reaction). Mark the beyond-the-data questions as ideas for discussion in Lesson 10 only. Reinforce: 'We use only the data from our clip; the new questions help us think about what else we would need to measure.'	Driving Question Board Tracking: moving from questions to evidence-based answers shows learners their own progress and keeps the whole class oriented toward the driving question. Sorting new questions identifies the limits of the model.	Observe whether sticky notes give correct values with units and cite the source (graph, table or calculation). Check that at least one question from each group links to safety or relative speed. Indicators of progress: learners who previously asked only 'how fast?' now ask about margins, time and conditions.

for readings Search terms: linear motion overtaking relative				
Model Building Phase  VIDEO: Interpreting direction of motion from velocity-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: linear motion overtaking relative	Individually, learners add a 'Safety Decision' panel to their class model of the overtake. The panel contains: (1) a labelled road diagram at $t = 0$ and $t = 7.5$ s showing A, B and L with the distances 150 m, 200 m, 150 m; (2) a small velocity-time graph for B (trapezium and rectangle, area 200 m) and A (horizontal line at 20 m/s, area 150 m); (3) a one-line calculation chain: $200 + 150 = 350$ m; $500 - 350 = 150$ m; $150 / 50 = 3.0$ s; (4) the what-if result (400 m gives 50 m, 1.0 s, unsafe); (5) a sentence stating the safety rule they have developed: 'An overtake is safe only when the distance to the oncoming vehicle is greater than the road used by the overtaking car and the oncoming vehicle, with a good time margin.' They compare their panel with a partner and improve it. Then they complete the Summary Table prompt in their books.	Say: 'Your model must let someone who was not in our class decide if Wanjiru can safely overtake. What must they see?' Give 10 minutes of silent individual work, then 5 minutes of partner critique with the prompt: 'Find one thing that is clear and one thing that could be clearer.' WAIT TIME: 15 seconds when asking for share-outs; cold-call 3 learners to read their safety rule. Highlight good models on the board. Close with an exit ticket: 'A lorry is 450 m away at $t = 0$. Using the same overtake (350 m used, closing speed 50 m/s), find the margin in metres and seconds, and say if you would overtake.' (Answer: 100 m, 2.0 s; learners justify their decision using the evidence.)	Cumulative Model Revision with Peer Critique: adding a decision panel connects graphs, calculations and diagrams into one coherent explanation, and peer critique helps learners refine their reasoning and communicate it clearly. The exit ticket tests transfer to a new number.	Model checklist: correct distances (150 m, 200 m, 150 m); 350 m total; margin 150 m and 3.0 s; what-if 50 m and 1.0 s; labelled axes and scales (1 cm = 1 s on x-axis); a safety rule in words. Exit ticket indicators: correct $450 - 350 = 100$ m; $100 / 50 = 2.0$ s; a justified decision referring to the margin. Learners who miss the exit ticket receive a short follow-up with the evidence table at the start of Lesson 10.

D. TEACHER REFLECTION

After the lesson, consider: Which learners could chain the three tools (acceleration, area, relative speed) without prompts, and which needed the diagram? Did learners confuse adding distances of B and L (opposite directions) with subtracting distances of B and A (same direction)? Did anyone mix up m/s and km/h in the closing speed? Was the 3.0 s margin meaningful to learners, and did they appreciate why 1.0 s is unsafe? Were all learners able to take part in the discussions, including quieter ones? What evidence from the exit ticket will you use to plan the review at the start of Lesson 10, where learners present the final model and answer the driving question and the two KICD key inquiry questions?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What we observed: In the Naivasha clip, overtaking car B took 7.5 s to pass overtaken car A. In that time B travelled 200 m, A travelled 150 m and oncoming lorry L travelled 150 m. B gained 50 m on A.
What did I learn?	What we learned: We can combine acceleration ($a = 2 \text{ m/s}^2$), areas under velocity-time graphs and relative speed to decide if an

	overtake is safe. B and L together use 350 m of road. The safety margin is the distance to the lorry minus 350 m, and dividing by the closing speed of 50 m/s gives the margin in seconds.
How does this explain the phenomenon?	How this explains the phenomenon: With the lorry 500 m away, the margin is 150 m, or 3.0 s, so Wanjiru's overtake is safe. If the lorry were only 400 m away, the margin would be 50 m, or 1.0 s, which is unsafe. Both cars can have high speedometer readings, but safety depends on relative speeds and the road available, not on speedometer readings alone.

LESSON 10 (80 minutes (2 x 40-minute periods)): Final Model and Answer to the Driving Question: Overtaking on the Naivasha Highway

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners synthesise everything from Lessons 1-9 into one complete model of the overtaking event (annotated graphs plus calculations), present it, and use it to answer the driving question and the two KICD key inquiry questions. No new skill is taught.
Knowledge	Learners will be able to: (1) explain that velocity is the rate of change of displacement with time, a vector with direction shown by sign (Car A +20 m/s, oncoming lorry L -20 m/s); (2) explain that acceleration is the rate of change of velocity, $a = (v - u)/t$, so overtaking car B has $a = (30 - 20)/5 = 2 \text{ m/s}^2$; (3) use the area under the velocity-time graph to show B travels 200 m and A travels 150 m in 7.5 s; (4) use relative speed (10 m/s for B relative to A, 50 m/s closing speed for B and L) to explain why A looks slow from B; (5) use the 350 m of road needed and the lorry 500 m away to show a 150 m (3.0 s) safety margin.
Skills	Science and Engineering Practices: Developing and using models (completing, presenting and revising the final model); Constructing explanations (writing the Final Explanation using claim-evidence-reasoning); Analysing and interpreting data (checking the model against the split-time table); Engaging in argument from evidence (peer critique of models); Using mathematics and computational thinking (calculations of velocity, acceleration, area and relative speed). Core competencies: critical thinking and problem solving, communication and collaboration.
Attitudes	Learners appreciate the role of mathematics in keeping Kenyan road users safe, show respect for peers' ideas during model critique, display honesty in comparing the Lesson 1 predictions with the final model, and value responsible road use (Values: responsibility, integrity, respect).
Key Inquiry Question	1. How do you determine the velocity of a moving body? 2. How do you determine the acceleration of a moving body? Driving question: How can we use distance, time, speed, velocity and graphs to describe and predict how one car overtakes another on a Kenyan road?
Purpose in Storyline	This is the culminating lesson. In Lesson 1 learners saw the overtaking clip near Naivasha and were puzzled that both speedometers looked high yet A looked slow from B. Lessons 2-9 gave them tools: definitions (L2), velocity and acceleration calculations (L3), displacement-time graphs (L4-5), velocity-time graphs and areas (L6-7), relative speed (L8) and the safe-overtake decision (L9). Here they pull all of it together, compare with their Lesson 1 model, close out the Driving Question Board and answer the driving question.
Safety Notes	Classroom-based lesson; keep walkways clear during gallery walks and presentations. Do not encourage learners to imitate overtaking or measure speeds on a real road. Emphasise that real overtaking is dangerous and that the numbers show why drivers must leave a large margin, especially when oncoming vehicles such as lorry L are near.

B. LESSON OVERVIEW

Learners return to the Naivasha highway clip where Wanjiru in white Subaru B overtakes Mr Kamau's blue Toyota Axio A while Mr Otieno's lorry L approaches. In groups they complete a final model made of annotated velocity-time and displacement-time graphs, the split-time table and key calculations: B accelerates from 20 to 30 m/s in 5 s ($a = 2 \text{ m/s}^2$); B travels 200 m while A travels 150 m in 7.5 s; B gains 50 m (20 m gap + 5 m + 5 m + 20 m gap); B relative to A is only 10 m/s (36 km/h) while B and L close at 50 m/s (180 km/h); the lorry started 500 m away and B and L cover 350 m together, leaving 150 m or 3.0 s. Learners compare this with their Lesson 1 prediction model, present to peers, complete the Driving Question Board by marking each question answered with evidence, and write a Final Explanation answering the driving question and the two KICD questions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Position vs. time graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Velocity-Time Graphs (Flexbook) Source: CK-12  Search ARES for similar readings	Learners re-watch the overtaking clip (Nairobi-Nakuru highway near Naivasha) and, individually, take out their Lesson 1 prediction (how long and how far the overtake takes). On a half sheet they write a new prediction: 'If lorry L were as far away as in the clip, would Wanjiru's overtake be safe, and what numbers would prove it?' They then pair up and compare their Lesson 1 sketch with what they now think, and mark one idea they are keeping and one they would change.	Play the clip twice with sound off. Say: 'Look back at your Lesson 1 prediction. Now, before we open any notes, write what numbers you would need to decide if Wanjiru's overtake is safe.' WAIT TIME 10-15 seconds in silence before pairs talk. Cold-call 4 learners: 'What is one thing you predicted in Lesson 1 that you can now check with a number?' Record responses on the board under 'Then' and leave space for 'Now'. Do not confirm answers yet; say: 'Hold that thought, we will test it with our evidence.'	Prediction revisited (Think-Pair-Share): returning to a recorded prediction makes learners aware of how their thinking has changed and gives a concrete baseline for the later L1-versus-final model comparison.	Look for: learners naming quantities needed (time, distance travelled by B and A, relative speed, distance to lorry L); learners referencing specific Lesson 1 statements; learners who still say 'A looks slow because it is slow' (flag for support during the Observe phase). Tally on the clipboard how many of 4 cold-calls mention distance to the lorry.
Observe Phase  VIDEO: Position vs. time graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Velocity-Time Graphs (Flexbook) Source: CK-12  Search ARES for similar readings	Groups of 4 gather their evidence from Lessons 1-9: the split-time table ($t = 0, 1, 2, 3, 4, 5, 6, 7, 7.5$ s with B distance, A distance and B minus A), their velocity-time graph of B (trapezium 0-5 s, rectangle 5-7.5 s) and A (horizontal line at 20 m/s), and their relative speed and safety margin calculations. Using A3 paper and graph paper (1 cm = 1 s on the x-axis), they build a final model with four panels: (1) velocity-time graphs of A and B with gradient 2 m/s^2 labelled and areas $125 \text{ m} + 75 = 200 \text{ m}$ for B and $20 \times 7.5 = 150 \text{ m}$ for A; (2) a scale road diagram showing the 20 m gap, 5 m car lengths and 20 m finishing gap, totalling 50 m gained by 7.5 s (level at $t = 5$ s, B's rear passes A's front at 5.5 s); (3) relative speed panel (B relative to A rising from 0 to $10 \text{ m/s} = 36 \text{ km/h}$; B and L closing at $30 + 20 = 50 \text{ m/s} = 180 \text{ km/h}$; lorry velocity - 20 m/s); (4) safety panel ($200 + 150 = 350 \text{ m}$, $500 - 350 = 150 \text{ m}$,	Circulate with a checklist. Say: 'Every number on your model must be traceable to the table or to a graph. Show me where 200 m comes from.' Ask groups: 'What does the gradient of this line tell us? What does the area tell us?' WAIT TIME 10-15 seconds after each question before prompting. Cold-call 3 groups to point to the table row ($t = 5$ s: B 125 m, A 100 m, gained 25 m) that matches their milestone. Prompt groups missing a direction sign: 'Which direction is positive? What sign does lorry L get?' Do not give answers; redirect to Lesson 3 (velocity and acceleration calculations) and Lesson 7 (gradient and area) work.	Evidence mapping with a consensus model: learners assemble multiple representations (table, graph, diagram, calculation) of the same event and must reconcile them, which builds understanding that these are different views of one motion.	Check: gradient labelled 2 m/s^2 with working $(30 - 20)/5$; areas correctly split into trapezium $(20+30)/2 \times 5 = 125 \text{ m}$ and rectangle $30 \times 2.5 = 75 \text{ m}$; A = 150 m; 50 m gained decomposed as $20 + 5 + 5 + 20$; lorry velocity negative; units (m/s, km/h, m, s) present. Note groups with arithmetic or sign errors and give a targeted question, not the answer.

	150/50 = 3.0 s). Each group checks that every number matches the split-time table.			
Explain Phase  VIDEO: Acceleration vs. time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Average velocity and average speed from graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each group presents its model in a 3-minute gallery presentation (one speaker, one pointer, one timekeeper, one recorder) while other groups rotate. Presenters must explain why Car A looks slow from Car B even though both speedometers show high values (72 km/h and up to 108 km/h), and why the manoeuvre is safe in the clip but would not be if lorry L were 400 m away. Visitors use a critique card: 'One thing the model explains well; one question; one suggestion.' Afterwards learners individually write the Final Explanation in claim-evidence-reasoning form: Claim answering the driving question; Evidence using at least four numbers from the model; Reasoning linking velocity (rate of change of displacement), acceleration (rate of change of velocity), graphs (gradient and area) and relative speed. They also write short answers to the two KICD questions: how to determine velocity (displacement divided by time, or gradient of a displacement-time graph) and how to determine acceleration (($v - u$)/t, or gradient of a velocity-time graph).	Set norms: 'We critique the model, not the people.' During presentations say: 'Presenters, tell us the viewpoint when you give a relative speed: B relative to A, or B and L together.' After each rotation ask: 'Which part of the model did you find most convincing, and why?' WAIT TIME 10-15 seconds before accepting responses. Cold-call 5 learners across groups to restate in their own words why A looks slow from B (relative speed only 10 m/s in the same direction, starting from 0 when acceleration begins) compared with the 50 m/s closing speed with L. For the what-if, ask: 'If lorry L were 400 m away, what margin remains?' Expect 400 - 350 = 50 m, 1.0 s, unsafe. Provide the CER sentence frames to learners who need them: 'I claim... The evidence is... This shows that...'	Claim-Evidence-Reasoning with peer argumentation: requiring numerical evidence and explicit reasoning, then having peers challenge it, pushes learners from describing what happened to explaining why, and uses their model as an argument.	Rubric (0-3) for the Final Explanation: correct claim; at least four accurate numbers ($a = 2 \text{ m/s}^2$, 200 m vs 150 m, 7.5 s, 10 m/s, 50 m/s, 150 m / 3.0 s); reasoning that links relative speed to the appearance of slow motion; correct KICD answers (velocity = displacement/time; $a = (v - u)/t$). Observable errors to flag: using the sum instead of the difference of speeds for same-direction motion; confusing distance and displacement; forgetting that the road must be shared with L (350 m). Collect exit sheets to target follow-up.
Driving Question Board (DQB) Creation  VIDEO: Interpreting change in speed from velocity-time graph	Learners return to the Driving Question Board started in Lesson 1 (questions such as 'How long does overtaking take?', 'How far do the cars travel?', 'Why does A look slow from B?', 'Is it safe with the lorry coming?'). In their groups they take each question card, decide whether it is Answered,	Say: 'Take one question from Lesson 1. Show the class the evidence that answers it.' WAIT TIME 10-15 seconds, then cold-call 6 learners (one per question category: time, distance, speed, velocity/acceleration, relative speed, safety). Ask: 'Which question could we not have	Question-tracking (DQB closure): linking each original question to specific evidence shows learners the arc of their learning and legitimises remaining questions as next steps.	Observe: each note cites specific evidence (e.g., 'Lesson 7: area under B's graph = 200 m'); learners correctly classify questions; at least one new, testable question per group. Record how many of the 6 cold-calls give an evidence-based response without teacher

<i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Instantaneous velocity and instantaneous speed from graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Partly answered or Still open, and write on a sticky note the evidence (a number, graph or lesson) that answers it, then place the note on the board. They add new questions that arise, for example: 'What if the road were wet?', 'What if Car A also sped up?' or 'What changes with a slope or a bend?'. The class reads the completed board and decides the final answer to the driving question.	answered without graphs? Which needed relative speed?' For open questions, say: 'Good questions we cannot yet answer are the start of new science; we leave them here for future lessons, for example how road conditions would change these numbers.' Do not add new content; note that wet-road braking and bends need ideas outside this sub-strand.		prompting.
Model Building Phase  VIDEO: Interpreting direction of motion from velocity-time graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Calculating Acceleration from Velocity and Time (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES	Learners compare their Lesson 1 model (sketch/prediction with no numbers) with the final model side by side using a two-column table: 'Lesson 1 model' and 'Final model' with rows for time, distance, speed/velocity, acceleration, relative speed and safety. They highlight what changed and what stayed the same, then revise any part of their final model that still has an error after peer critique. Each learner writes a short Summary Table entry (what we observed, learned, explained) and a two-sentence answer to the driving question, for example: 'Using distance, time, velocity, acceleration and graphs, we find that B, accelerating at 2 m/s^2 to 30 m/s, takes 7.5 s and 200 m to overtake A, which travels 150 m at 20 m/s; because B relative to A is only 10 m/s, A looks slow; and since B and lorry L need 350 m of the 500 m, a 150 m (3.0 s) margin	Say: 'Place your Lesson 1 model next to your final model. What can your final model do that your first could not?' WAIT TIME 10-15 seconds before sharing. Cold-call 4 learners to name one change and the evidence that caused it. Challenge: 'If Mr Kamau in Car A also sped up during the overtake, which part of your model would change?' Take 2 volunteers. Close by reading the driving question aloud and asking the class to give the one-sentence answer in chorus, then invite 3 learners to read their road-safety messages. Praise honest revision: 'Changing your model because of evidence is exactly what scientists do.'	Model comparison and revision (then-versus-now): explicit side-by-side comparison makes conceptual change visible and lets learners see mathematics as a tool to describe and predict, not just compute.	Look for: Lesson 1 model correctly characterised (qualitative, no numbers or mixed up speed and relative speed); final model includes graphs with gradient and area, correct figures and units; the driving-question answer includes velocity, acceleration, relative speed and the safety margin; road-safety message is consistent with the data (for example, leave a large margin because closing speed with an oncoming lorry is $50 \text{ m/s} = 180 \text{ km/h}$). Use a 4-level rubric (beginning, developing, secure, extending).

for similar readings	remains.' Finally, learners connect to their own lives by writing one road-safety message for a Kenyan driver or boda boda rider based on the numbers.			
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D. TEACHER REFLECTION

After the lesson, reflect: Which parts of the final model did learners complete without help, and which numbers (for example 350 m or 3.0 s) caused most errors? Did cold-calling reveal learners still thinking that A looks slow because it is slow? Compare Lesson 1 predictions with final explanations for at least five learners: how many now use relative speed correctly and state the viewpoint? Were the 10-15 second wait times observed, and did quieter learners contribute? Which open DQB questions could be the basis for the next sub-strand or an extension project? Note adjustments for the next class, including extra practice on gradient and area, and whether the Kenyan contexts (Naivasha highway, matatus, boda boda) supported engagement.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed Car B (white Subaru) overtake Car A (blue Toyota Axio) near Naivasha: both start at 72 km/h (20 m/s), B speeds up to 108 km/h (30 m/s) in 5 s and is back in lane at 7.5 s, while the oncoming lorry L approaches at 20 m/s.
What did I learn?	We learned that velocity = displacement/time and $a = (v - u)/t$; that the gradient of a velocity-time graph gives acceleration (2 m/s^2) and the area gives distance (B 200 m, A 150 m); and that relative speed is the difference of speeds in the same direction (10 m/s) and the sum in opposite directions (50 m/s).
How does this explain the phenomenon?	We explained that A looks slow from B because B relative to A is only 10 m/s (36 km/h), and that the overtake is safe because B and L need 350 m of road, leaving 150 m (3.0 s) of the 500 m; with the lorry at 400 m the margin is only 1.0 s, which is unsafe.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.